Tournaments not inducible by five voters

In loving memory of Adam Paton.

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Abstract

How many members must a committee have before its pairwise majorities can express any collective preference? With enough voters every pattern of pairwise majorities, or tournament, can arise, but a committee of k voters produces only some; for odd k, let $N(k)$ be the least number of alternatives on which some tournament arises from no k voters. Only $N(3) = 8$ is known (Shepardson and Tovey 2009; Eggermont et al. 2013). For five voters the best bounds were $11 \leq N(5)$ (Bachmeier et al. 2019) and $N(5) \leq 41$ (Eggermont et al. 2013, p. 271; Bachmeier et al. 2019), and the smallest explicit tournament known to escape five voters had over 6×10^8 vertices, from a theorem of Fidler (2011) (Bachmeier et al. 2019). We show $13 \leq N(5) \leq 22$. The upper bound comes from nine tournaments on 22 vertices that five voters cannot produce: eight one-vertex deletions of doubly regular tournaments on 23 vertices, including every deletion of one of them, and one arc reversal. The Paley tournament on 23 vertices, which Bachmeier et al. (2019) left open after six weeks of SAT solving, is beyond five voters, and so is every other doubly regular tournament on 23 vertices. The lower bound comes from an analysis of all tournaments on 12 vertices. Along the way, counting profiles against near-regular tournaments gives $N(5) \leq 38$ and improves the counting bound for every odd $k \leq 21$. An argument that searches no profiles shows that the Paley tournament on 43 vertices is also beyond five voters. Its predictability, the largest share of the voters that a profile can give to every majority at once, exceeds the threshold 3/5 by only 2/1505, so this result refutes the conjecture that a predictability above the threshold lets five voters produce a tournament. Both doubly regular tournaments on 19 vertices, the Paley tournament and one other, arise from five voters, but not with unit margin, that is, not with every majority carried by three votes to two. Our method is a search that places one alternative at a time and exploits the tournament's symmetries. We certify four of the negative results with machine-checkable proofs, and a public repository reproduces every result.

Keywords: majority voting; tournaments; McGarvey's theorem; Paley tournaments; constraint search; SAT

1. Introduction

Let A be a finite set of alternatives. A *profile* is a finite list of voters, each of whom ranks A by a linear order. When the number of voters is odd, the *majority tournament* of the profile has, for each pair of alternatives u and v, the arc $u \to v$ if more than half of the voters rank u above v, and the arc $v \to u$ otherwise. McGarvey (1953) proved that every tournament on A, that is, every choice of one arc for each pair, is the majority tournament of some profile. So when the number of voters is unrestricted, majority rule imposes no structure on collective preference: every pattern of pairwise majorities occurs, including patterns with cycles. McGarvey's construction uses two voters for each pair of alternatives, so the number of voters it needs grows with the number of alternatives, and how many voters a pattern needs is our question.

Once the number of voters is fixed, majority rule does impose structure. A panel of k voters admits only finitely many profiles, so it can produce only finitely many patterns, and once the number of alternatives is large enough most patterns are beyond it. A pattern that some k-voter panel does produce is called *k-inducible*. The question is then one about small deliberative bodies: on an agenda of n alternatives, is a committee of k members expressive enough to realize every collective preference one could write down, or are there patterns of majority preference that k members can never produce, however they rank the alternatives?

For three voters the answer is known. Every pattern on at most seven alternatives is realizable, and on eight alternatives one is not, by the work of Shepardson and Tovey (2009) and of Eggermont, Hurkens and Woeginger (2013). We write $N(k)$ for the least number of alternatives at which some pattern escapes k voters, so that $N(3) = 8$. For five voters, the smallest open case, the answer was known only to lie between 11 (Bachmeier et al. 2019) and 41 (Eggermont et al. 2013, p. 271; Bachmeier et al. 2019), and the smallest pattern known to escape five voters had over 6×10^8 alternatives, from a theorem of Fidler (2011) (Bachmeier et al. 2019).

This paper narrows that to $13 \leq N(5) \leq 22$. Five voters suffice on every agenda of at most twelve alternatives: we show this for all $\approx 9.0 \times 10^8$ patterns on eleven alternatives together with all of their one-alternative extensions, and our code produces an explicit five-voter profile for any one of them. And on twenty-two alternatives there are patterns that five voters cannot produce. We find eight of them, each obtained by removing one alternative from a perfectly balanced pattern on twenty-three alternatives (a *doubly regular tournament*, in the terminology defined below). The most symmetric of them consists of two copies of the Paley tournament on eleven alternatives, joined by a ladder (Figure 1); the Paley tournaments form an infinite family, with one tournament for every prime number of alternatives congruent to 3 modulo 4. Two arguments that search no profiles come first: counting profiles shows that some pattern on thirty-eight alternatives escapes five voters, without exhibiting one, and a short argument shows that the Paley tournament on forty-three alternatives does. Our search then settles the Paley tournament P_{23} on twenty-three alternatives, the instance that a general-purpose Boolean satisfiability (SAT) solver failed to decide in six weeks of computation (Bachmeier et al. 2019), in twenty-two hours on a laptop, and goes on to the patterns on twenty-two alternatives.

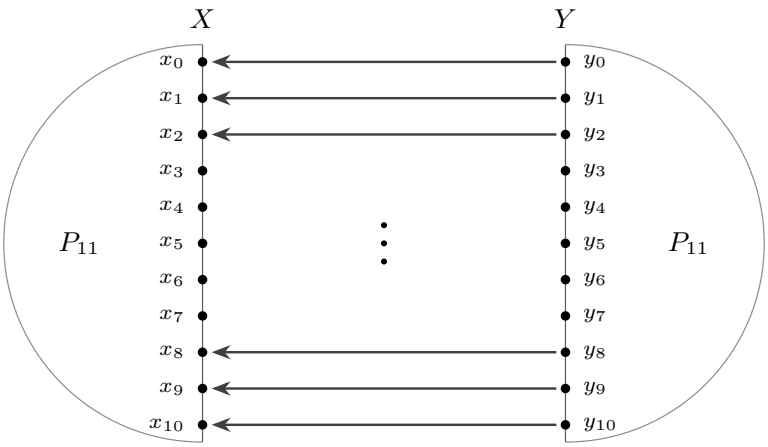


Figure 1: The tournament $\Delta_1 - v_{10}$ (notation of Section 1.3), which is not 5-inducible. Its 22 vertices split into $X = \{x_0, \ldots, x_{10}\}$ and $Y = \{y_0, \ldots, y_{10}\}$, with indices taken modulo 11, and each half induces a copy of P_{11}: $x_i \to x_j$ exactly when $j - i$ is a non-zero square modulo 11, that is, lies in $\mathrm{QR} = \{1, 3, 4, 5, 9\}$, and $y_i \to y_j$ exactly when $j - i$ is a non-square. Between the halves, $x_i \to y_j$ when $j - i$ is a non-square, and $y_j \to x_i$ otherwise, that is, when $j - i$ lies in $\mathrm{QR} \cup \{0\}$. In particular every rung $y_i \to x_i$ of the ladder points from Y to X; only six rungs are drawn. Each x_i therefore has out-degree 10 and each y_j has out-degree 11. Adding 1 to every index, and multiplying every index by an element of QR, are automorphisms; together they generate the full automorphism group, of order 55.

The smallest Paley tournament that five voters cannot produce, P_{23}, is as balanced as a collective preference on twenty-three alternatives can be: every alternative defeats exactly eleven others, and every pair jointly defeats the same number. It is the least decisive pattern available, in the sense that no alternative and no pair is distinguished from any other. What these *obstructions* (tournaments that are not 5-inducible) share is balance, not symmetry: P_{23} is also highly symmetric, but the other 36 doubly regular tournaments on twenty-three alternatives are just as balanced, 32 of them have no symmetry at all, and all of them escape five voters too (Section 6.7).

Two further findings concern the margins by which majorities are carried. A five-member panel can carry a pairwise majority by three votes to two, by four to one, or unanimously, and it is natural to ask whether the bare three-to-two margins suffice. They do not: P_{19} is realizable by five voters, but every profile realizing it must carry some majority by more than the bare minimum (Section 6.4). And the failures are fragile in a precise sense: reversing a single pairwise majority in P_{19} or in P_{23} restores realizability (Sections 6.3 and 6.6).

We use the definitions of our companion paper (Chindelevitch and Harutyunyan 2026), in the notation below. Let T be a *tournament* on the vertex set V, $|V| = n$: a digraph in which every unordered pair $\{i, j\}$ carries exactly one of the arcs $i \to j$ or $j \to i$. A *k-profile* consists of k linear orders $\pi_1, \ldots, \pi_k$ on V. The *majority tournament* of such a profile puts an arc from i to j whenever $|\{v : i \prec_{\pi_v} j\}| > k/2$. For odd k this is well-defined on every pair, and the result is a tournament. We say T is *k-inducible* if some k-profile has T as its majority tournament, and we call such a profile a *witness* for T. The *converse* of T is the tournament obtained from T by reversing the direction of every arc.

We call the number of voters who rank i above j the *support* of the arc $e : i \to j$, and denote it $c(e)$. If $c(e) > k/2$, the *margin* of e is $2c(e) - k$. In a witness every arc's support exceeds $k/2$, so for $k = 5$ it is 3, 4 or 5. A witness has *margin at most M* if every arc has margin at most M. For $k = 5$ the possible margins are 1, which we call *unit margin*, 3 (every support in $\{3, 4\}$), and 5 (no restriction, i.e. plain majority). A unit-margin witness is in particular a majority witness. In our companion paper we conjectured that the margin-based hierarchy is strict, which for $k = 5$ asserts two separate things: that some tournament is 5-inducible with margin at most 3 but not with unit margin, and that some tournament is 5-inducible but not with margin at most 3. We settle the first here, since P_{19} is such a tournament (Section 6.4), while the second remains open (Section 7.2).

Reversing every voter's order reverses every arc of the majority tournament and changes no support. So a profile realizes T exactly when its reversal realizes the converse $\bar{T}$, at the same margins, and T and $\bar{T}$ are k-inducible for the same k and at the same margins. Every question in this paper is therefore unchanged by taking the converse. Accordingly, we count a family enumerated in full, such as all tournaments of a given order or all vertex-transitive ones, up to isomorphism, and a family that our searches generated up to isomorphism and converse.

One further quantity recurs below. Informally, the *predictability* $\alpha^*(T)$ is the largest fraction $\alpha > 1/2$ for which there is some profile, on any number t of voters, such that every arc of T is supported by at least αt of them. Since every arc of a k-voter witness is supported by more than $k/2$ of the voters, k-inducibility forces $\alpha^*(T) \geq (k + 1)/2k$, which is $2/3$ at $k = 3$ and $3/5$ at $k = 5$. The implication is only one-directional: α^* can meet the threshold and the tournament still fail to be inducible. Shepardson and Tovey (2009) introduced the notion. The converse, that $\alpha^*(T) \geq (k + 1)/2k$ implies that T is k-inducible, is implicit in their work, which left open, at a referee's suggestion, whether every tournament on seven vertices is 3-inducible; we call it the *threshold conjecture*. Our companion paper refutes it at $k = 3$, where it fails only when α^* equals the threshold exactly. At $k = 5$ it fails strictly above the threshold, for P_{23}, with $\alpha^* = 7/11$ (Section 6.1), and for P_{43}, where α^* exceeds $3/5$ by only $2/1505$ (Section 4).

Counting arguments show that almost all tournaments fail to be k-inducible for any fixed k, yet the *explicit* examples they yield are astronomically large. This paper closes much of the gap for $k = 5$. Let $N(k)$ be the least order of a tournament that is not k-inducible. The answer is only known exactly for $k = 3$, where $N(3) = 8$ (Shepardson and Tovey 2009; Eggermont et al. 2013). For $k = 5$ the state of the art before this work was $11 \leq N(5) \leq 41$, the lower bound due to Bachmeier et al. (2019) and the upper bound to Eggermont et al. (2013, p. 271) and Bachmeier et al. (2019). We first raise the lower bound to 12 by an exhaustive census of all $\approx 9.0 \times 10^8$ tournaments on 11 vertices, all of which are 5-inducible (Appendix A.1), and then to 13.

The upper bounds in the literature are of two kinds, and they are not always comparable. Some are non-constructive, and obtained by counting: one compares the number of tournaments of a given order with the number that k linear orders can produce, and when the first exceeds the second, some tournament of that order must be missed. Eggermont et al. (2013, p. 271) and Bachmeier et al. (2019) argued this way to reach 41, and restricting the comparison to regular tournaments, whose out-degrees are all equal, and near-regular ones, whose out-degrees are as nearly equal as an even order allows, reaches 38 (Section 3). Other bounds are constructive, and exhibit an explicit tournament. The previous record for $k = 5$ had over 6×10^8 vertices,

from a theorem of Fidler (2011) (Bachmeier et al. 2019). We first show that P_{43} is not 5-inducible (Section 4), the first explicit example of moderate order. Our search then shows that P_{23} is not 5-inducible, which improves both the best non-constructive (counting) and the best constructive (explicit) upper bounds to $N(5) \leq 23$, and then that eight tournaments on 22 vertices are not 5-inducible either, so that $N(5) \leq 22$.

1.1 Special properties of Paley tournaments

The Paley tournament P_q, for a prime $q \equiv 3 \pmod 4$, has vertex set $\mathbb{F}_q$ and an arc from i to j whenever $j - i$ is a nonzero square, or *quadratic residue*; we write QR for the set of these, so that $i \to j$ exactly when $j - i \in \mathrm{QR}$. As everywhere in this paper, vertices are numbered from 1, vertex $i + 1$ being the field element i. It is *doubly regular*: every vertex has out-degree $(q - 1)/2$ and every ordered pair of vertices has exactly $(q - 3)/4$ common out-neighbours. Its automorphism group $\mathrm{Aut}(P_q)$ (Section 2) has order $q(q - 1)/2$. This makes P_q *arc-transitive*, meaning that for every pair of arcs e, e' of P_q there is an automorphism of P_q that takes e into e'. It is also *self-converse*, meaning that it is isomorphic to its own converse; the relevant mapping is multiplication by -1, which is easily checked to not be an automorphism when $q \equiv 3 \pmod 4$.

1.2 Neither SAT nor the slack argument settles Paley(23)

Paley tournaments are natural candidates for non-inducibility because they are far from being transitive, and because their symmetry makes an otherwise hopeless search merely very hard. Bachmeier et al. (2019) call them *quadratic residue tournaments* and write Q_q; their orientation is the converse of ours, which changes no result here (Section 1); in particular a tournament and its converse have the same *majority dimension*, the least k for which a tournament is k-inducible. We use the name Paley and the notation P_q throughout.

The SAT study of Bachmeier et al. (2019) establishes

$$\dim(Q_3) = \dim(Q_7) = 3, \qquad \dim(Q_{11}) = \dim(Q_{19}) = 5,$$

where dim denotes majority dimension, and then stops exactly where the problem becomes hard:

> "Unfortunately, we were not able to check whether the majority dimension of Q_{23} is equal to 5 or larger as the SAT solver did not terminate within a total of six weeks."

Nor does the argument by which we settle P_{43} (Section 4), which searches no profiles. It rests on the *slack*, the amount by which five voters can collectively fall short of the most arcs one order can agree with; the slack is proportional to $\alpha^* - 3/5$, and it is 6 at $q = 43$ but 46 at $q = 23$ (Table 2), which leaves the voters of a candidate profile for P_{23} far too much room.

Predictability separates $k = 5$ from $k = 3$. At $k = 3$ predictability decides exactly at $N(3)$: the 96 non-3-inducible tournaments on eight vertices have $\alpha^* = 13/20$, which is *below* $2/3$, so the predictability condition detects them, while all other tournaments on eight vertices or fewer with at least one cycle have $\alpha^* = 2/3$ (Shepardson and Tovey 2009); with the computation of Eggermont et al. (2013), $N(3) = 8$ follows. At $k = 5$ it does not: Table 2 has $\alpha^* > 3/5$ at every q, yet P_{23}, P_{31} and P_{43} are all non-inducible, with P_{23} the farthest *above* the threshold among the three.

1.3 Contributions

1. **There are non-5-inducible tournaments on 22 vertices, so $N(5) \leq 22$** (Section 6.2). We write Δ_i for the i-th of the 37 doubly regular tournaments on 23 vertices in the catalogue of McKay (2026), with vertices $v_1, \ldots, v_{23}$, and $\Delta_i - v_j$ for Δ_i with vertex v_j deleted. Eight such tournaments on 22 vertices, pairwise non-isomorphic and none the converse of another, are not 5-inducible, and all eight are vertex-critical (deleting any vertex leaves a 5-inducible tournament). Three of them are the three one-vertex deletions of Δ_1 up to isomorphism. One of these, $\Delta_1 - v_{10}$, is two copies of P_{11} joined by a ladder, and a symmetry reduction (Corollary 5.5) makes its refutation cheap; reversing one of its arcs gives a ninth obstruction (Section 6.3). $\Delta_{11} - v_{20}$ also consists of two copies of P_{11}, joined differently. The refutations of $\Delta_1 - v_{10}$ and $\Delta_6 - v_5$ are formally certified (Appendix B).

2. P_{23} **is not** 5**-inducible** (Section 6.1). This settles the instance on which the SAT solver of Bachmeier et al. (2019) did not terminate within a cumulative six weeks, and was the first step from 38 down to 23. We also decide P_{31}, which does not improve the bound; the search settles it in less time than P_{23}.

3. **Counting gives** $N(5) \leq 38$**, and** P_{43} **is not** 5**-inducible** (Sections 3 and 4). Four voters and an out-degree sequence determine the tournament of a fifth (Proposition 3.1), which sharpens the counting bound to $N(5) \leq 38$ and improves the counting bounds of Bachmeier et al. (2019) for every odd $k \leq 21$ (Theorem 3.3). The predictability of P_{43} exceeds 3/5 by only 2/1505; a lemma on how the voters can rank the arcs of a cyclic triangle and an exhaustive screen of about 1.7 billion orders then show that it is not 5-inducible (Theorem 4.1), which refutes the threshold conjecture at $k = 5$ strictly above the threshold.

4. P_{19} **is** 5**-inducible, but not** 5**-inducible with unit margin** (Section 6.4). This is an example suggesting that $k = 5$ may qualitatively differ from $k = 3$. For $k = 3$, the first k-inducible tournaments that are non-k-inducible with unit margin appear after $N(k)$, order 9 vs order 8 (see our companion paper). For $k = 5$ they appear no later than order 19, so the order is reversed unless $N(5) \leq 19$, and the search of Appendix C gives evidence against that.

5. P_{19} **and** P_{23} **are arc-critical** (Sections 6.3 and 6.6). Reversing an arc in P_{19} restores 5-inducibility at unit margin, and reversing an arc in P_{23} restores unrestricted 5-inducibility.

6. P_{31} **and** P_{43} **are not vertex-critical** (Section 6.5). Deleting a vertex from P_{31} or P_{43} leaves a tournament that is still not 5-inducible.

7. **Four formally certified negative results** (Appendix B). P_{19} is not 5-inducible at unit margin, and P_{23}, $\Delta_1 - v_{10}$ and $\Delta_6 - v_5$ are not 5-inducible. The search is split into independent subproblems (*cubes*); each is refuted by the SAT solver CaDiCaL, which emits its refutation as a machine-checkable *LRAT* proof (Section 1.4); and every one of those proofs is then rechecked by a separate program that shares no code with the solver.

8. **Every tournament on** 12 **vertices is** 5**-inducible, even at unit margin** (Appendix A). The analysis settles all 2,048 one-vertex extensions of each of the $\approx 9.0 \times 10^8$ tournaments on 11 vertices, all of which are 5-inducible (Appendix A.1); it is complete because every 12-vertex tournament is such an extension of each of its twelve one-vertex deletions.

1.4 Related work

This paper builds on two others. Our companion paper (Chindelevitch and Harutyunyan 2026) settles the three-voter questions from which this one starts, and we use its definitions. Bachmeier et al. (2019) is the prior art for both the *bounds* we improve and the *method*: it gave $11 \leq N(5) \leq 41$, the upper bound also due to Eggermont et al. (2013, p. 271), and an explicit non-5-inducible tournament on over 6×10^8 vertices, from a theorem of Fidler (2011), and it brought SAT to bear on k-majority digraphs, characterized the 3-majority case building on Dushnik and Miller (1941), and established hardness results for several voting rules with a constant number of voters.

The value of $N(3)$ is settled. Shepardson and Tovey (2009) introduced the predictability $\alpha^*(T)$ defined above. A 3-voter profile is in particular a 2/3-supermajority representation, so $\alpha^*(T) < 2/3$ rules out 3-inducibility; their 8-vertex tournament has $\alpha^* = 13/20$, which gives $N(3) \leq 8$. They also prove that every tournament on at most 7 vertices has $\alpha^* \in \{2/3, 1\}$, and Eggermont, Hurkens and Woeginger (2013) then show by direct computation using an integer linear programming (ILP) formulation that every tournament on 7 vertices is 3-inducible, and every tournament on 8 or 9 vertices is 5-inducible, yielding $N(3) = 8$ and $N(5) \geq 10$.

Our certified computation (Appendix B) is standard. It splits the problem into independent subproblems, using the *cube-and-conquer* method (Heule et al. 2012): a SAT instance is split by fixing a few variables at a time, giving many partial assignments called *cubes*, and each cube is handed to the solver as a problem of its own. We adopt the standard of evidence set by the Boolean Pythagorean triples proof (Heule et al. 2016), later re-checked with a formally verified checker (Cruz-Filipe et al. 2019), in which every subproblem emits a machine-checkable refutation that an independent checker validates. Our refutations are in the LRAT format

(Cruz-Filipe et al. 2017) (Appendix B.1), a format in which the solver records each clause it derives so that an independent checker can replay and confirm the entire derivation.

The rest of this paper is organized as follows. Section 2 sets out the notation. Section 3 bounds $N(k)$ by counting, and Section 4 proves that P_{43} is not 5-inducible by an argument that searches no profiles. Section 5 describes the search: the symmetry reductions on which it relies and its five steps. Section 6 presents the results of the search: P_{23} and P_{31} (Section 6.1), the eight 22-vertex obstructions that give $N(5) \leq 22$ (Section 6.2), vertex- and arc-criticality (Sections 6.3 to 6.6), and other structured families (Section 6.7). Section 7 discusses where the method may transfer and what remains open, and Section 8 describes the reproducibility material.

2. Preliminaries

In this section we fix the notation used throughout.

Throughout, T is a tournament on $V = \{1, \ldots, n\}$ and $k = 5$ unless stated otherwise. We write $N^+(v)$ and $N^-(v)$ for the out- and in-neighbourhoods of v. A tournament is *regular* if $|N^+(v)| = |N^-(v)|$ for each $v \in V$; this can only happen for n odd. For even n it is *near-regular* if every out-degree is $n/2 - 1$ or $n/2$; since the out-degrees sum to $\binom{n}{2}$, exactly $n/2$ vertices then have each of the two. Following our companion paper, D_n, R_n and S_n denote the numbers of isomorphism classes of, respectively, all tournaments, regular tournaments and self-converse tournaments on n vertices. The text gives them to two significant figures and Appendix D tabulates their exact values. For any $S \subseteq V$, the sub-tournament of T induced by S, written $T[S]$, is the tournament obtained by restricting T to the vertices in S and the arcs between those vertices.

We call a tournament T *vertex-critical* for k-inducibility at margin M if T is not k-inducible at margin M, but $T - v$ is k-inducible at margin M for every vertex v. Similarly, we call T *arc-critical* for such a property if T has it, but any T' obtained from T by reversing an arc does not.

An *automorphism* of a tournament T is a permutation of its vertices carrying every arc to an arc; these form a group $\mathrm{Aut}(T)$ under composition. The identity is always an automorphism; any other automorphism is *non-trivial*, and T is *rigid* when it has none. For a permutation group Γ on a set X, the *orbit* of x is $\Gamma x = \{\gamma(x) : \gamma \in \Gamma\}$, and the orbits partition X. The group is *transitive* on X when there is a single orbit, that is, when for any $x, x' \in X$ there is a $g \in \Gamma$ with $g(x) = x'$. We use orbits of $\mathrm{Aut}(T)$ on the vertices, on the arcs, and on the *ordered pairs* of distinct vertices, where γ sends (u, v) to $(\gamma(u), \gamma(v))$. An ordered pair is either an arc or the reverse of one, so that third action has at least two orbits.

The next section uses this notation to bound $N(k)$ by counting.

3. Counting bounds on $N(k)$

In this section we bound $N(k)$ from above by counting, without exhibiting a tournament: five voters cannot produce every pattern of majorities on 38 alternatives, and the counting bounds of Bachmeier et al. (2019) improve for every odd $k \leq 21$.

Comparing counts of profiles and tournaments gives $N(5) \leq 41$ non-constructively (Eggermont et al. 2013, p. 271; Bachmeier et al. 2019). Restricting the count to *regular* tournaments tightens this to $N(5) \leq 39$, and to *near-regular* tournaments to $N(5) \leq 38$, because four voters already determine a tournament of prescribed out-degrees.

3.1 Four voters determine a tournament with given out-degrees

The fifth voter of a profile is constrained only on the pairs that the other four split evenly, and there a prescribed out-degree sequence leaves it no choice.

Proposition 3.1 (algorithmic completion uniqueness). *Fix odd k, linear orders $\pi_1, \ldots, \pi_{k-1}$ on $[n]$, and a target out-degree sequence $d\colon [n] \to \mathbb{Z}_{\geq 0}$. There is at most one tournament with out-degree sequence d realized by adding a k-th voter to the profile; if it exists, it can be constructed in $O(n^2)$ time.*

Proof. Among the pairs $\{x, y\}$, those split with margin ≥ 2 by $\pi_1, \ldots, \pi_{k-1}$ are majority-decided regardless of π_k; the evenly split $(\frac{k-1}{2}\text{–}\frac{k-1}{2})$ pairs form an undirected graph D, and on them the majority agrees with π_k. The completed tournament is therefore determined by the orientation R that π_k induces on D – an acyclic orientation, being the restriction of a linear order – and conversely every acyclic orientation of D is induced by some π_k (extend it to a topological order). Prescribing d determines, at each vertex x, the number $b(x)$ of D-edges that R must orient out of x. The following algorithm constructs the unique candidate:

1. Set $b(x) := d(x) - |\{\text{majority-decided arcs leaving } x\}|$ for every vertex. If some $b(x) < 0$, or $\sum_x b(x) \neq |E(D)|$, halt: **no completion exists**.
2. While D has surviving vertices: if no surviving vertex has $b(x) = 0$, halt: **no completion exists**. Otherwise pick such an x, orient every surviving D-edge $\{x, y\}$ from y to x, decrement $b(y)$ for each such edge (halting likewise if some $b(y)$ drops below 0), and delete x.
3. Output the tournament formed by the majority-decided arcs together with the constructed orientation R; a suitable order π_k is the reverse of the deletion order, which is a topological order of R (this is exactly Kahn's (1962) topological sorting algorithm).

Every step is forced: in any acyclic orientation of the surviving graph there is a sink, and a sink uses none of its out-budget, so a vertex with $b(x) = 0$ exists whenever a valid orientation does; conversely a vertex with $b(x) = 0$ can orient no surviving edge outward, so all of its edges are forced inward, and deleting it updates its neighbours' budgets correctly. Hence if the algorithm succeeds, its output is the unique orientation realizing d, and if it halts early, no completion exists. The output is acyclic: every arc points from a later-deleted to an earlier-deleted vertex. Each vertex is processed once and each edge oriented once, for $O(n^2)$ time in total. ■

3.2 Counting near-regular tournaments gives $N(5) \leq 38$

For even n there are no regular tournaments, but the near-regular ones substitute for them (Section 2). Deleting any vertex v from a regular tournament on $n+1$ vertices leaves a near-regular tournament on n: the $\frac{n}{2}$ in-neighbours of v each lose an out-arc, dropping to out-degree $\frac{n}{2} - 1$, while the $\frac{n}{2}$ out-neighbours keep out-degree $\frac{n}{2}$. Conversely, each near-regular tournament on n arises from a unique such regular tournament: re-add v beating exactly the $\frac{n}{2}$ vertices of out-degree $\frac{n}{2} - 1$. Hence this is a bijection and

$$\#\{\text{near-regular tournaments on } n\} \;=\; R^L_{n+1}. \tag{3.1}$$

Here and below, R^L_ν denotes the number of *labelled* regular tournaments on ν vertices – the counting argument takes place in the labelled world – while R_n without the superscript always denotes the unlabelled (isomorphism) count of Section 2.

Theorem 3.2. *Some near-regular tournament on 38 vertices is not 5-inducible; hence $N(5) \leq 38$.*

Proof. Dropping one voter from a realization shows every 5-inducible tournament on $[n]$ with a prescribed out-degree sequence is the unique (Proposition 3.1) completion of some 4-multiset of orders; distinct such inducible tournaments determine distinct (4-multiset, out-degree sequence) pairs. Fixing the *precise* out-degree sequence is exactly what makes the completion unique, so the count is one per fixed sequence: the near-regular sequences on even n number $\binom{n}{n/2}$ (the defining choice is which half of the vertices have out-degree $n/2$), so

$$\log_2 \#\{\text{5-inducible near-regular on } n\} \;\leq\; \log_2\Big[\binom{n}{n/2}\binom{n!+3}{4}\Big] \;<\; 624.7 \quad (n = 38).$$

Equivalently, by vertex-relabelling symmetry each of the $\binom{n}{n/2}$ sequences is realized by $R^L_{n+1}/\binom{n}{n/2}$ tournaments, which must exceed $\binom{n!+3}{4}$ for some of them not to be 5-inducible. By (3.1) their total is R^L_{39}, and Schrijver's (1983) lower bound on Eulerian orientations at K_{39} (a regular tournament is an Eulerian orientation of K_{n+1}) gives

$$\log_2 \#\{\text{near-regular on } 38\} \;=\; \log_2 R^L_{39} \;\geq\; \log_2\Big[2^{-\binom{39}{2}}\binom{38}{19}^{39}\Big] \;>\; 625.5.$$

Since $625.5 > 624.7$, some near-regular 38-tournament is not 5-inducible. (The plain regular count already gives the weaker $N(5) \leq 39$: $\log_2 R^L_{39} > 625.5 > 610.8 > \log_2 \binom{39!+3}{4}$, so all but a $2^{-14.8}$ fraction of regular 39-tournaments are non-5-inducible.) ■

Remark. The same proof with $k = 3$ shows a pair admits at most one regular 3-voter majority completion, so some regular tournaments on 17 or more vertices are not 3-inducible, consistent with, and far weaker than, $N(3) = 8$ (Shepardson and Tovey 2009; Eggermont et al. 2013).

3.3 The counting bound improves for every odd $k \leq 21$

The argument is not specific to five voters: Proposition 3.1 holds for every odd k, and the near-regular bijection (3.1) does not depend on k. Counting both classes gives:

Theorem 3.3 (all odd k). *For odd $k \geq 3$: (a) at most $\binom{n!+k-2}{k-1}$ regular tournaments on odd $[n]$ are k-inducible, so a non-k-inducible regular tournament exists once $R_n^L > \binom{n!+k-2}{k-1}$; (b) at most $\binom{n}{n/2}\binom{n!+k-2}{k-1}$ near-regular tournaments on even $[n]$ are k-inducible, so a non-k-inducible near-regular tournament exists once $R_{n+1}^L > \binom{n}{n/2}\binom{n!+k-2}{k-1}$, using in both cases Schrijver's bound $R_\nu^L \geq 2^{-\binom{\nu}{2}}\binom{\nu-1}{(\nu-1)/2}^\nu$ (odd ν).*

Together with the exact multiset count of profiles, this extends Table 1 of Bachmeier et al. (2019); Table 1 lists both. Their counting lemma bounds the majoritarian expressiveness $n^{\mathcal{T}}(k)$, the largest n such that every n-vertex tournament is k-inducible, via the estimate $\binom{n!+k-1}{k} \leq (2 \cdot n!)^k / k!$. Using the exact multiset count in place of their estimate, which uses the bound of Fiol (1992) (the "exact" row), already sharpens three entries (the $k = 5$ case noted in a footnote of Bachmeier et al. 2019); the regular and near-regular bounds of Theorem 3.3 then sharpen all ten. The 10 best bounds are equally split between *regular* tournaments of odd size and *near-regular* tournaments of even size. The smallest non-k-inducible tournament has $N(k) = n^{\mathcal{T}}(k) + 1$ vertices, so that $n^{\mathcal{T}}(5) \leq 37$ is exactly the bound $N(5) \leq 38$ of Theorem 3.2.

Table 1: Upper bounds on the majoritarian expressiveness $n^{\mathcal{T}}(k)$, the largest n such that every n-vertex tournament is k-inducible. Columns are indexed by the odd voter count k; bold marks a strict improvement on Table 1 of Bachmeier et al. (2019).

	3	5	7	9	11	13	15	17	19	21
Bachmeier et al. (2019)	18	41	66	93	122	152	183	216	249	282
exact	18	**40**	**65**	93	122	152	183	**215**	249	282
Thm 3.3	**16**	**37**	**62**	**89**	**117**	**147**	**178**	**210**	**243**	**276**

Each threshold is sharp for the class attaining it: the count wins at the order $n^{\mathcal{T}}(k) + 1$ and falls short one vertex lower.

These bounds show that a non-5-inducible tournament exists on 38 vertices, but they do not exhibit one. The next section gives an explicit example on 43 vertices.

4. Paley(43) is not 5-inducible

No committee of five voters produces P_{43}, although its predictability exceeds the threshold 3/5. Section 3 showed that a non-5-inducible tournament exists on 38 vertices; in this section we prove that P_{43} is one, by an argument that searches no profiles. Among the Paley tournaments of Table 2, P_{43} is the only one that the margin of its predictability over 3/5 settles on its own: that margin is just 2/1505, and it leaves five voters too little room. P_{23}, with $\alpha^* = 7/11$, refutes the threshold conjecture further above the threshold (Section 6.1), but only through the search. We write $P = P_{43}$ throughout the section.

Theorem 4.1. *$P = P_{43}$ satisfies $\alpha^*(P) = \mathrm{MAS}/C = 543/903 = 181/301 > 3/5$, yet P is not 5-inducible. Hence $\alpha^* > \frac{k+1}{2k}$ does not imply k-inducibility, and $N(5) \leq 43$ with an explicit tournament.*

Here $\mathrm{MAS}(T)$ is the maximum number of arcs of T that a single linear order agrees with. An arc $u \to v$ is *forward* in an order π if π ranks u above v and *backward* otherwise, and $\mathrm{fwd}_T(\pi)$ is the set of forward arcs. The identity $\alpha^* = \mathrm{MAS}/C$ holds because P is arc-transitive, that is, $\mathrm{Aut}(P)$ acts transitively on its arcs. No order has more than MAS forward arcs, so in any profile the average support of an arc is at most a fraction MAS/C of the voters, and the smallest support is no larger. Conversely, the profile formed by the images

under $\mathrm{Aut}(P)$ of one order with MAS forward arcs gives every arc exactly that fraction (the symmetrization lemma of our companion paper). Throughout, $q = 43$, $C = \binom{q}{2} = 903$, the number of cyclic triangles is $N_\Delta = (q^3 - q)/24 = 3311$, and $|\mathrm{Aut}(P)| = q(q-1)/2 = 903$.

4.1 The slack forces two near-optimal voters

In a witness every arc is forward for at least three of the five voters, so the voters have at least $3C$ forward arcs between them, while none has more than MAS. The *slack*

$$\mathrm{slack}_5(T) \;=\; 5\,\mathrm{MAS}(T) - 3C$$

therefore bounds how far the voters can collectively fall short of the maximum. For an arc-transitive T it equals $5C\,(\alpha^*(T) - 3/5)$, so it grows in proportion to C whenever the predictability stands off from the threshold. A subset dynamic program modulo $\mathrm{Aut}(P)$ with a meet-in-the-middle step (Appendix E) gives $\mathrm{MAS}(P) = 543$, so $\mathrm{slack}_5(P) = 2715 - 2709 = 6$, the smallest value in Table 2; the others range from 7 to 47.

For an inducing profile $(\pi_1, \ldots, \pi_5)$, let $f_i = |\mathrm{fwd}_P(\pi_i)|$ and relabel the voters so that $f_1 \geq f_2 \geq \cdots \geq f_5$. Then $\sum_{i=1}^{5} f_i \geq 3C = 2709$ and $f_1 \leq 543$, so the other four sum to at least 2166, and $4f_2 \geq 2166$, i.e. $f_2 \geq 542$. The *level* of an order π is $\delta(\pi) = \mathrm{MAS} - |\mathrm{fwd}_P(\pi)|$, so **at least two voters lie at level** ≤ 1, and the five levels sum to at most $\mathrm{slack}_5(P) = 6$.

A cyclic triangle is *double-backed* by an order that ranks two of its three arcs backward, and the *double-back set* $\mathrm{DB}(\pi)$ of π is the set of cyclic triangles it double-backs.

Lemma 4.2 (co-backing). *In any inducing profile of 5 voters, every cyclic triangle is double-backed by at most one voter.*

Proof. Each arc of the triangle is ranked backward by ≤ 2 of the 5 voters, so there are at most six backward incidences; each voter ranks ≥ 1 arc of a cyclic triangle backward (a linear order cannot agree with all three arcs of a cycle); if d voters double-back, incidences $= 5 + d \leq 6$, so $d \leq 1$. ■

The five voters' double-back sets are therefore pairwise disjoint, and the two voters at level ≤ 1 form a pair of orders of level ≤ 1 with disjoint double-back sets. **If no such pair exists,** P **is not 5-inducible**, and this is a finite criterion.

4.2 No two near-optimal orders have disjoint double-back sets

The orders of level at most 1, which we call the *shell*, number 1,662,696,609, in 1,841,303 orbits of $\mathrm{Aut}(P)$ (Appendix F). We decide the criterion by checking every pair, which two reductions make practical:

- **A window of 24 vertices.** We fix the window $\mathcal{W} = \{0, 1, \ldots, 23\}$ and its 538 cyclic triangles. The restriction of $\mathrm{DB}(\pi)$ to these triangles depends only on how π ranks $\mathcal{W}$, so the shell has only 4,709,640 distinct restrictions, and two orders with disjoint double-back sets have disjoint restrictions; only the 5,092,111 disjoint pairs of restrictions need the exact check.
- **Orbit representatives on one side.** $\mathrm{DB}(\sigma\pi) = \sigma(\mathrm{DB}(\pi))$ for $\sigma \in \mathrm{Aut}(P)$, where $\sigma\pi$ ranks $\sigma(v)$ where π ranks v, because σ maps forward arcs of π to forward arcs of $\sigma\pi$. Overlap sizes are therefore Aut-invariant, and it suffices to test one representative of each orbit against all shell orders, which saves a factor of 903.

The exact check compared all 4,376,325,129 pairs of an orbit representative and a shell order whose restrictions are disjoint, and **none has disjoint double-back sets**; the smallest overlap is 68 triangles. With the co-backing lemma, this proves Theorem 4.1. ■

Table 2: The Paley family: size, MAS, predictability, slack and status.

q	C	MAS	α^*	$\approx$	slack_5	5-inducible?
7	21	14	2/3	0.6667	7	yes; 3 voters suffice
11	55	35	7/11	0.6364	10	yes; 5 voters needed
19	171	107	107/171	0.6257	22	yes; 5 voters needed
23	253	161	7/11	0.6364	46	no (Thm 6.2)
31	465	285	19/31	0.6129	30	no (Sec. 6.1)
43	903	543	181/301	0.6013	**6**	**no (Thm 4.1)**
47	1081	658	14/23	0.6087	47	open

The table shows that α^* is not monotone in q: it coincides at $q = 11$ and $q = 23$, and is larger at $q = 47$ than at $q = 43$, so extrapolating within the family is unreliable. The slack is small only at $q = 43$, so the proof above exploits a numerical coincidence. P_{11} and P_{19} are 5-inducible, so non-5-inducibility is not generic to the family; the slack argument does not reach P_{23} or P_{31}, which the search of Section 6 decides, and P_{47} remains open.

Whether $\alpha^* > \frac{k+1}{2k}$ implies k-inducibility for odd $k \geq 7$ remains open. Adding an order and its reverse to a profile changes no margin, so k-inducibility implies $(k+2)$-inducibility, and a counterexample at $k = 7$ must be non-5-inducible. P_{43} meets the $k = 7$ threshold ($181/301 > 4/7$), but both the arithmetic of the slack and the co-backing lemma fail at $k = 7$. Theorem 3.3 guarantees non-7-inducible tournaments on 63 vertices, without exhibiting one.

Before this work, the smallest explicit non-5-inducible tournament had over 6×10^8 vertices, from a theorem of Fidler (2011) (Bachmeier et al. 2019). The slack argument reaches no further than P_{43}, so the next section describes the search, which settles P_{23} and P_{31} and finds the obstructions on 22 vertices.

5. The search

In this section we describe the search that decides the instances of Section 6: the symmetry reductions on which it relies (Section 5.1) and the search itself, in five steps (Section 5.2). Every search starts from a small set B of vertices, the *base*, and a *base state* of B: k linear orders of B whose majorities agree with $T[B]$, counted up to relabelling the voters (Appendix G.2).

5.1 Symmetry reductions

Two symmetries act on witnesses, and every method here uses both: the $k!$ relabellings of the voters, which form the symmetric group S_k, and $\text{Aut}(T)$ acting on the vertices. A permutation σ of the voters acts on a profile $\pi = (\pi_1, \ldots, \pi_k)$ by reordering it, and a permutation h of V acts on a linear order by relabelling, so that $h\pi_j$ ranks $h(u)$ above $h(w)$ exactly when π_j ranks u above w, and on a profile voter by voter. The two actions commute, so the group $S_k \times \text{Aut}(T)$ acts on profiles. It maps witnesses to witnesses: permuting voters does not change any margin, and the margin of $h(u) \to h(w)$ after relabelling by an automorphism h equals that of $u \to w$ before. Every reduction we use is a special case of one principle.

Theorem 5.1 (orbit principle). *Let G be a subgroup of $S_k \times \text{Aut}(T)$ acting on a set Ω, and let φ map profiles to Ω with $\varphi(g\pi) = g\,\varphi(\pi)$ for every $g \in G$. Let R be a subset of Ω. Then if R contains an element of every G-orbit and T is k-inducible, T has a witness π with $\varphi(\pi) \in R$.*

Proof. Let w be a witness and choose $g \in G$ with $g\,\varphi(w) \in R$. Then gw is a witness and $\varphi(gw) = g\,\varphi(w) \in R$. ■

We never apply Theorem 5.1 directly. What the methods use are its corollaries below; the theorem gives them a common proof and tells us when two of them can be combined.

The corollaries below differ only in the group G and in the object $\varphi(\pi)$ that a profile is reduced to (Table 3). The group is often a subgroup because the object dictates it: $\text{Aut}(T)$ acts on the base states of B only through the automorphisms that map B onto itself.

Table 3: The symmetry reductions of this section as instances of Theorem 5.1. Here B is the base and H is a subgroup of $\mathrm{Aut}(T)$ mapping a subset $C \subseteq B$ onto itself.

result	group G	object $\varphi(\pi)$
Corollary 5.2	$\mathrm{Aut}(T)$	the top two vertices of π_1
Corollary 5.3	$\mathrm{Aut}(T)$	the top vertex of π_1
Corollary 5.4	S_k	the profile π itself
Corollary 5.5	$S_k \times H$, $h(C) = C$	the base state of $C \subseteq B$ that π extends

As a witness can be transformed by an automorphism, we may in some cases decide in advance what some voter's top two vertices are.

Corollary 5.2 (orbit anchoring). *Let* $\mathrm{Aut}(T)$ *act on the ordered pairs of distinct vertices, and choose one representative pair from each orbit, say* $(a_1, b_1), \ldots, (a_m, b_m)$*. If* T *is* k*-inducible, then* T *has a witness in which, for at least one index* i*, some voter ranks* a_i *first and* b_i *second.*

Proof. Theorem 5.1 with $G = \mathrm{Aut}(T)$, $\varphi(\pi)$ the ordered pair of the top two vertices of π_1, and R the chosen pairs. ■

Thus we may search only for witnesses whose top pair is one of the m chosen pairs. The corollary only guarantees that *some* i works, not any particular one.

For a Paley tournament $m = 2$, and the representatives can be exhibited without computing $\mathrm{Aut}(P_q)$. For $a \in \mathrm{QR}$ and any t, the map $x \mapsto a(x - t)$ multiplies every difference by a, which cannot change whether a difference lies in QR, so it is an automorphism; it sends the ordered pair (t, s) to $(0, a(s - t))$, and as a ranges over QR, $a(s - t)$ ranges over the coset $\mathrm{QR} \cdot (s - t)$. Since $\mathbb{F}_q^*$ is the disjoint union of QR and its complement, each of them one such coset, these maps alone leave only two orbits, separated by whether the difference lies in QR. The orbits of the full group are unions of these, so there are at most two, and since automorphisms carry arcs to arcs, at least two. They are therefore exactly the arcs and the non-arcs, and any arc together with any non-arc serves as representatives. On P_7, worked through in Figure G.1, $\mathrm{QR} = \{1, 2, 4\}$, so the representatives are $(1, 2)$, whose difference 1 lies in QR, and $(1, 4)$, whose difference 3 does not; both lie inside the base $B = \{1, 2, 4\}$ used there, which is what lets the anchoring be checked from the base state alone, and panel (c) rejects an insertion that would lift a vertex above an anchored pair.

Corollary 5.3 (vertex anchoring). *Let* $\mathrm{Aut}(T)$ *act on the vertices and choose one representative* $v_1, \ldots, v_r$ *from each orbit. If* T *is* k*-inducible, then* T *has a witness in which some voter ranks some* v_i *first.*

Proof. Theorem 5.1 with $G = \mathrm{Aut}(T)$, $\varphi(\pi)$ the top vertex of π_1, and R the chosen vertices. ■

In both corollaries a *representative* is one of the chosen vertices or ordered pairs of vertices, one from each orbit of $\mathrm{Aut}(T)$; it is part of T, not a base state. Corollaries 5.2 and 5.3 apply only if the representatives meet every orbit, so we compute $\mathrm{Aut}(T)$ and its orbits for every tournament, with `nauty` 2.8.6 (McKay and Piperno 2014), which also provides every automorphism group, orbit set and canonical form reported in this paper. We say that a witness meeting either restriction is *anchored*. The two are alternatives rather than a sequence, each without loss of generality, and which is better depends on the tournament's number of orbits on vertex pairs compared with that on vertices, as well as the relative complexity of testing each restriction.

Corollary 5.4 (voter lex-ordering). *If* T *is* k*-inducible, then* T *has a witness whose* k *orders are non-decreasing under any fixed total order on linear orders.*

Proof. Theorem 5.1 with $G = S_k$, φ the identity, and R the profiles whose orders are non-decreasing; every S_k-orbit contains one, obtained by sorting. ■

Two reductions compose when the second does not undo the first. After restricting to witnesses with $\varphi_1(\pi) \in R_1$, a second application of Theorem 5.1 may use only elements of its group that preserve this property. Anchoring followed by sorting is such a pair: every permutation of the voters preserves the property Corollary 5.2 guarantees, "*some* voter ranks a_i first and b_i second", which depends on the multiset of orders

and not on which voter carries it. Had it been sharpened to "voter 1 ranks a_i first and b_i second", the composition would fail. Both are therefore available to every formulation in Appendix G.

Together they reduce the running time substantially, in two ways. Corollary 5.4 is absorbed into the definition of a base state: the k orderings of B are required to be non-decreasing, dividing the count by almost 120 (almost, because a tuple with two equal orders is fixed by some of the 120 permutations).

Corollary 5.2 acts earlier still, *invalidating* base states outright, but only when *all* the representative pairs lie inside B. Suppose they do, and that a base state extends to an anchored witness, some voter of which ranks a_i first and b_i second overall. As $a_i, b_i \in B$, that voter ranks a_i first and b_i second *within* B, which the base state alone reveals. Contrapositively, a base state no voter of which tops B with a chosen pair extends to no anchored witness, hence to no witness at all, and may be discarded. If one representative has an endpoint outside B this filter does not apply; a base state failing the test on the representatives it sees may still extend to a witness anchored on the one it cannot. We call a base state *live* if it survives. On the five-vertex base we use on the Paley tournaments 2,591 of the 8,031 are live.

The search and the certification differ in *when* the condition in Corollary 5.2 is tested. Both test it at the start to determine the live base states. The search keeps testing it throughout, since a base state fixes the orders only on B whereas the condition constrains the full orders: a voter picking (a_i, b_i) as its top pair in B stops being anchored once a later vertex is inserted above b_i. Pruning is sound once every representative has both endpoints placed, for then the test is monotone. An insertion can destroy an anchor but never create one, the placed vertices' relative order being already fixed, so a partial profile with no anchored voter has no extension with one. The SAT certification instead tests the condition once, at the root, where it decides which cubes reach the solver.

Corollaries 5.2 and 5.3 restrict, without loss of generality, what a witness looks like on all of V, and so how the base interacts with the vertices outside it. When a subgroup of automorphisms maps the base onto itself, the same idea works inside the base: the subgroup identifies base states with one another, and only one from each class needs to be searched. Every witness $\pi_1, \ldots, \pi_k$ determines a base state of B, namely the multiset of the restricted orders $\pi_1|_B, \ldots, \pi_k|_B$ (Appendix G.2 gives the full definition), and we say that the witness *extends* this base state. The *restriction* of a base state of B to a subset $C \subseteq B$ is the multiset of the restrictions of its k orders to C, which is a base state of C.

Corollary 5.5 (base-stabilizer quotient). *Let H be a subgroup of $\mathrm{Aut}(T)$ and $C \subseteq B$ with $h(C) = C$ for every $h \in H$ (possibly $C = B$), and choose one base state of C from each H-orbit. Call a base state of B chosen if its restriction to C is chosen.*

(a) H permutes the base states of C, and if one of them extends to a witness, so does every base state in its H-orbit. Consequently, if T is k-inducible, some chosen base state of B extends to a witness.

(b) Suppose in addition that every element of H fixes each representative vertex of Corollary 5.3. Then, if T is k-inducible, some chosen base state extends to a witness anchored in the sense of Corollary 5.3.

(c) Suppose instead that every element of H fixes both a_i and b_i for every representative pair (a_i, b_i) of Corollary 5.2. Then, if T is k-inducible, some chosen base state extends to a witness anchored in the sense of Corollary 5.2.

Proof. As $h(C) = C$, relabelling by h maps an order of C to an order of C, and since h restricts to an automorphism of $T[C]$, it maps a base state of C to a base state of C; this is the action of H, and S_k acts trivially, base states being multisets. The base state of C that $h\pi$ extends is h applied to the one π extends, so Theorem 5.1 with $G = S_k \times H$ and $\varphi(\pi)$ that base state gives a witness extending a chosen base state Q_C of C; applying h^{-1} gives the orbit statement. The restrictions $\pi_1|_B, \ldots, \pi_k|_B$ of that witness form a base state Q_B of B that it extends, and the restriction of Q_B to C is Q_C, so Q_B is chosen. For (b), an element of H maps a voter ranking v_i first to one ranking $h(v_i) = v_i$ first, so it preserves anchoring, and the composition rule above applies. For (c), likewise, an element of H maps a voter ranking a_i first and b_i second to one ranking $h(a_i) = a_i$ first and $h(b_i) = b_i$ second. ■

With $C = B$, the search therefore needs only one base state of B from each H-orbit, which divides the work by up to $|H|$, and Corollary 5.4 still holds because base states remain multisets of orders. The condition on

anchoring cannot be dropped: if H moves some representative, mapping an anchored witness into the chosen base state can destroy its anchor, and only one of the two reductions can be used.

Finding H is a question about $\mathrm{Aut}(T)$. Automorphism groups of tournaments have odd order, since an automorphism of order 2 would swap the two endpoints of some arc. By Cauchy's theorem, $\mathrm{Aut}(T)$ contains an element of order p for every prime p dividing its order, and an element h maps B onto itself exactly when B is a union of cycles of h. When $|B| = 5$ and h has order 5, a single 5-cycle of h is such a base, and the subgroup $H = \{\mathrm{id}, h, h^2, h^3, h^4\}$ of its powers divides the number of base states by up to 5. The base states fixed by h are the only ones whose orbit is smaller than $|H|$.

A subgroup that maps a base of the size we search with onto itself need not exist, and this is where $C \subsetneq B$ helps. An automorphism of order 3 with no fixed point, for example, maps no set of 5 vertices onto itself, but it maps each of its 3-cycles onto itself. With C one such 3-cycle inside a base B of 5 vertices, the search keeps only the base states of B above a chosen base state of C. As H need not map B onto itself, it does not act on the base states of B, and the fraction kept is not exactly $1/|H|$; we choose in each orbit the base state of C with the fewest base states of B above it. Three of the unit-margin searches of Appendix C used this form, keeping 600 of the 2,200 base states of a five-vertex base inducing the regular tournament on five vertices.

Using the whole group. Anchoring and the quotient use different parts of $\mathrm{Aut}(T)$, in the way the orbit–stabilizer theorem (Dummit and Foote 2004, Section 4.1, Proposition 2) describes. Anchoring (Corollaries 5.2 and 5.3) uses the orbits of the chosen representatives, and the quotient (Corollary 5.5) uses a subgroup that fixes every representative. We therefore take $H = G_R$, the largest such subgroup, which meets the condition of Corollary 5.5(b) or (c) by construction. For a single representative r, let $\mathrm{Aut}(T) \cdot r = \{h(r) : h \in \mathrm{Aut}(T)\}$ be its orbit and $G_r = \{h \in \mathrm{Aut}(T) : h(r) = r\}$ its stabilizer; then $|\mathrm{Aut}(T)| = |\mathrm{Aut}(T) \cdot r| \cdot |G_r|$, so between them the two reductions involve the whole group: its transitive part through anchoring and the stabilizer through the quotient. With several representatives, G_R is the intersection of their stabilizers and can be smaller, so we choose the representatives to keep it large, and then choose the base among the unions of cycles of G_R, or bases containing such a union (Corollary 5.5 with $C \subsetneq B$).

The two parts do not save work at the same rate. The quotient divides the number of base states by up to $|G_R|$, whereas anchoring discards base states only when every representative lies in the base, and otherwise prunes during the search by an amount that depends on the tournament. Table 4 shows the split for the tournaments of Section 6. For the Paley tournaments the stabilizer of a representative pair is trivial, so the whole group goes to anchoring and the saving is well below $|\mathrm{Aut}(T)|$; for $\Delta_1 - v_{10}$ the group splits into orbits of 11 for anchoring and a stabilizer of order 5 for the quotient.

Table 4: How the searches and certificates of Section 6 use the automorphism group. Anchoring uses the orbits of the representatives and the quotient their common stabilizer. In the $\Delta_1 - v_{10}$ search the representatives lie outside the base, so anchoring prunes during the search and the count of base states kept reflects the quotient alone.

tournament	$\lvert\mathrm{Aut}(T)\rvert$	anchoring	quotient	base states kept
P_{19}, unit, certificate	171	ordered pairs	none	22,876 of 142,251
P_{23}, search	253	ordered pairs	none	2,591 of 8,031
P_{23}, certificate	253	ordered pairs	none	343,896 of 3,414,729
$\Delta_1 - v_{10}$, search	55	x_0, y_0 (orbits of 11)	order 5	1,611 of 8,031
$\Delta_1 - v_1$, $\Delta_1 - v_2$	5	none	order 5	1,611 of 8,031
$\Delta_{11} - v_2$, $-v_5$, $-v_{20}$	5	none	order 5	1,611 of 8,031
$\Delta_6 - v_5$, $\Delta_{28} - v_5$	1	none	none	8,031 of 8,031

Figure 2 summarizes which of these reductions apply to a given tournament and base, and how they combine.

5.2 The search in five steps

The search extends base states. A base state fixes how the k voters rank B; the search inserts the remaining $n - |B|$ vertices into those rankings one at a time, backtracking as soon as the arcs already decided contradict

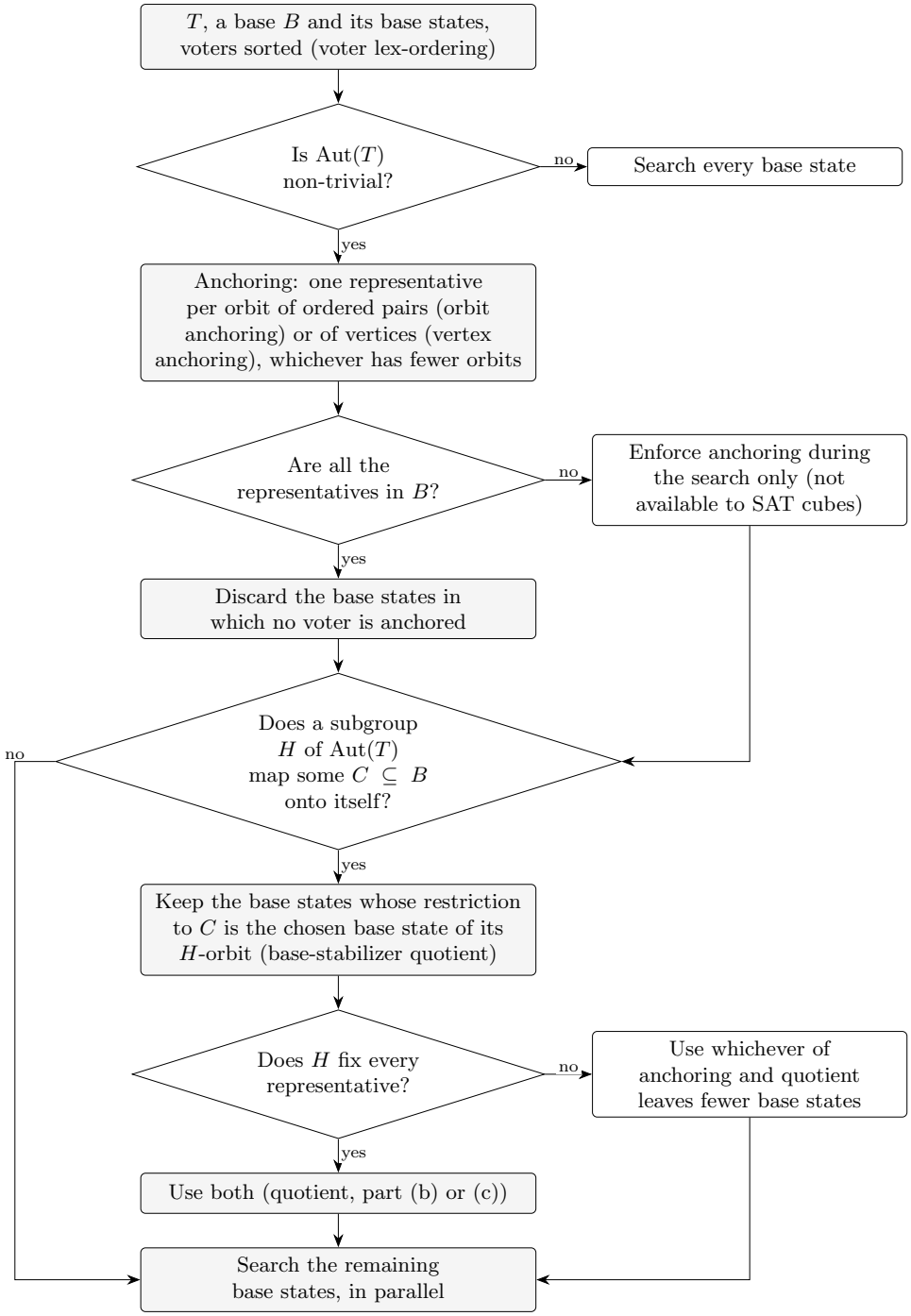


Figure 2: Which symmetry reductions apply to a tournament T and a base B. Anchoring by Corollary 5.2 or Corollary 5.3 can discard base states only when every representative lies in B, since a base state reveals the orders on B alone; otherwise the depth-first search enforces it while vertices are inserted, and the SAT route cannot use it. The quotient of Corollary 5.5 combines with anchoring only when H fixes every representative.

T. A base state extended to k full rankings yields a witness; one that cannot be is eliminated; and T fails to be k-inducible exactly when every base state is eliminated. Given T and the margin regime, the search has five steps:

1. **Pick a size.** Choose the base size b, in practice 5 or 6.
2. **Pick a subtournament.** Choose a set B of b vertices. Only $T[B]$ matters for what follows, so the choice can and should be optimized quickly (see Appendix G.4), since the base-state count varies by more than two orders of magnitude across the 12 isomorphism classes at $b = 5$.
3. **Find its base states.** Enumerate the k-tuples of linear orders of B whose supports agree with $T[B]$, counted up to voter permutation by Corollary 5.4.
4. **Determine the usable symmetries.** Compute $\mathrm{Aut}(T)$ with `nauty` and its orbits, then impose the ordered-pair anchoring of Corollary 5.2, the vertex anchoring of Corollary 5.3, or neither, discarding the base states no anchored witness can use. If a subgroup H maps B, or a subset C of it, onto itself, also keep only the base states above one representative of each H-orbit (Corollary 5.5), checking that H fixes the anchoring representatives (Figure 2).
5. **Run to completion, in parallel across base states.** The verdict is positive as soon as any base state yields a witness. It is negative only when *every* live base state has been searched to exhaustion.

Figure G.1 in Appendix G.3 shows all five steps on P_7 at $k = 3$. Steps 1–4 are fast and user-chosen; step 5 is a parallel computation. The same steps describe the SAT route, with step 5 replaced by handing each live base state to a conflict-driven clause-learning (CDCL) solver as a *cube* (the partial assignment fixing the voters' rankings of B), so the two share their decomposition and their symmetry reductions and differ only in what refutes a leaf in the search tree. The next section reports what this search decides.

6. Results

In this section we present our results, first for the Paley tournaments on 23 and 31 vertices and for the eight obstructions on 22 vertices, then for criticality and for the other structured families. Table B.2 in Appendix B.2 lists what each result rests on.

The two regimes we explore are *unrestricted* (or plain majority) inducibility, and *unit margin*. The former is the standard notion of inducibility, and the latter asks that every arc be carried by exactly three voters against two. Only an unrestricted refutation bounds $N(5)$. However, the lemma below shows that for the Paley family and most tournaments derived from it, the unrestricted inducibility is equivalent to inducibility with margin at most 3 (i.e. support at most 4).

We also use unit margin as a fast filter for unrestricted inducibility. A unit-margin witness is in particular a majority witness, and the unit-margin search is much faster (Appendix G.5), so when the question is only whether a tournament is 5-inducible we search at unit margin first and at plain majority only if no unit-margin witness exists. The order-12 analysis of Appendix A works this way, and so do several of the family searches on 21 vertices (Appendix C). A tournament that fails the filter but has a majority witness is then a unit-margin obstruction found as a by-product, and several of those reported below arose this way.

Lemma 6.1 (3-cycle bound). *Let $u \to v \to w \to u$ be a directed triangle of T and let $\pi_1, \dots, \pi_k$ be any linear orders with k odd. Then*

$$c(u \to v) + c(v \to w) + c(w \to u) \le 2k.$$

Consequently, in any majority witness, no arc lying in a directed triangle has unanimous support k.

Proof. A linear order cannot rank u above v, v above w and w above u simultaneously, so each voter agrees with at most two of the three arcs; summing over the k voters gives the inequality. If $c(u \to v) = k$, the other two supports sum to at most k, contradicting the majority requirement that each be at least $(k + 1)/2$. ■

For $k = 3$ the second conclusion is Proposition 1 of Milosz, Hamel and Pierrot (2018), with a transitivity argument; we need it for every odd k, and in the summed form above.

6.1 Paley(23) and Paley(31) are not 5-inducible

Theorem 6.2. *P_{23} is not 5-inducible. Consequently $N(5) \leq 23$.*

The proof is a search that runs to completion. The base is $\{1, 2, 3, 6, 12\}$, which induces the regular tournament on 5 vertices and gives 8,031 base states. They are reduced to 2,591 by Corollary 5.2, by requiring one voter to rank either $(1, 3)$ or $(1, 6)$ as the first pair. None of them produces a witness. This statement is also formally certified in Appendix B.1.

Read as a statement about voting, it says that a five-member panel cannot produce the collective preference P_{23} on twenty-three alternatives: no choice of five rankings, out of the $23!^5$ available, has P_{23} as its pairwise-majority relation. It is the first such statement for an agenda of fewer than forty-three alternatives, and the first for any tournament whose non-realizability a general-purpose solver could not settle.

The same method settles a larger case: P_{31} is not 5-inducible, over 21,009 base states with a different base, reduced to 4,007 by Corollary 5.2. Appendix G.5 shows how the runtime scales in q, and Appendix G.6 tabulates the cost of every computation in this paper. Settling P_{31} shows that our search decides 31-vertex instances on a single laptop within a day, which the SAT and ILP solvers we tried did not.

6.2 Eight tournaments on 22 vertices that five voters cannot produce

P_{23} is one of 37 doubly regular tournaments on 23 vertices, which the catalogue of McKay (2026) lists up to isomorphism. We write Δ_i for the i-th tournament in that list, with its vertices $v_1, \ldots, v_{23}$ numbered in the catalogue's own labelling, and $\Delta_i - v_j$ for the tournament left when vertex v_j is deleted. Every one of the 37 is beyond five voters, at unit margin and unrestricted alike (Section 6.7), so the doubly regular tournaments on 23 vertices are a natural place to look for a smaller obstruction one vertex down.

Theorem 6.3. *The tournaments $\Delta_1 - v_1$, $\Delta_1 - v_2$, $\Delta_1 - v_{10}$, $\Delta_6 - v_5$, $\Delta_{11} - v_2$, $\Delta_{11} - v_5$, $\Delta_{11} - v_{20}$ and $\Delta_{28} - v_5$, on 22 vertices, are not 5-inducible. In particular, no tournament obtained from Δ_1 by deleting one vertex is 5-inducible. Consequently $N(5) \leq 22$.*

The second statement follows from the first because the automorphism group of Δ_1, of order 55, has three vertex orbits, of sizes 11, 11 and 1, which contain v_1, v_2 and v_{10} respectively. The eight are pairwise non-isomorphic, and none is isomorphic to the converse of another. Each has out-degrees 10 and 11, eleven vertices of each, and every arc lies in at least five directed triangles, so by the 3-cycle bound no arc of a witness could be unanimous.

All eight share an outline inherited from the doubly regular tournament they come from. Deleting a vertex v of a doubly regular tournament on 23 vertices leaves its 11 out-neighbours and its 11 in-neighbours. Each set induces a regular tournament on 11 vertices, and the 121 arcs between them are such that every out-neighbour of v beats 6 in-neighbours and every in-neighbour beats 5 out-neighbours. The obstructions differ in the two halves and in how they are joined. In $\Delta_1 - v_{10}$ and $\Delta_{11} - v_{20}$ both halves are P_{11}. In $\Delta_1 - v_1$, $\Delta_1 - v_2$, $\Delta_{11} - v_2$ and $\Delta_{11} - v_5$ the two halves are again isomorphic to each other and to each other's converse, but each has an automorphism group of order 5 rather than 55. In $\Delta_{28} - v_5$ the two halves are converses of each other, each with no non-trivial automorphism. In $\Delta_6 - v_5$ the halves are neither isomorphic nor converse, and neither has a non-trivial automorphism. Appendix H gives the adjacency matrices of all eight.

The halves suggest a definition. We call a vertex v of a tournament T a *bisector* if $T[N^+(v)]$ is isomorphic to $T[N^-(v)]$ or to its converse, an *isomorphic* or a *converse* bisector respectively; a vertex can be both. A tournament is *bisectable* if it has a bisector or arises from a tournament with one by deleting a bisector, and *non-bisectable* otherwise. A bisector has equal in- and out-degree, so the first case needs odd order and the second even order, and a tournament of even order is bisectable exactly when its vertices split into two halves of equal size that induce isomorphic or converse tournaments. Bisectable tournaments are much easier to enumerate than the others, since one tournament on half the vertices fixes both halves.

Every doubly regular tournament on at most 23 vertices is bisectable, whereas 118 of the 722 on 27 vertices are not. On 23 vertices, P_{23} is self-converse and the other 36 form 18 pairs of converses. Every vertex is a bisector only in P_{23} and in Δ_1 and its converse Δ_{35}. The halves at a vertex of P_{23} are the same tournament at every vertex, and it is a half of no vertex of any other; in Δ_1 the halves at v_{10} are P_{11}, while those at v_1

and v_2 are not. Our obstructions cover every case. In $\Delta_1 - v_{10}$ and the five deletions with an automorphism of order 5 the deleted vertex is a bisector of both types, and in $\Delta_{28} - v_5$ a converse bisector, so these seven are bisectable; $\Delta_1 - v_{10}$ and $\Delta_{28} - v_5$ are bisectable through a single split, the one into their two halves. The deleted vertex of Δ_6 is not a bisector, and no other split works, so $\Delta_6 - v_5$ is non-bisectable.

The first is the most structured. Figure 1 shows $\Delta_1 - v_{10}$ as two copies of P_{11}, on $X = \{x_0, \ldots, x_{10}\}$ and $Y = \{y_0, \ldots, y_{10}\}$, joined by a ladder whose rungs all point from Y to X, and its automorphism group has order 55. Its two vertex orbits are X and Y. The automorphisms multiplying every index by a non-zero square fix x_0 and y_0 and form a subgroup H of order 5, which maps the five out-neighbours of x_0 in X, $B = \{x_1, x_3, x_4, x_5, x_9\}$, onto themselves. B induces the regular tournament on five vertices, so it has 8,031 base states, which H groups into 1,611 orbits. Corollary 5.5 therefore lets the search keep one base state per orbit, and since H fixes x_0 and y_0, it can at the same time anchor on them as the representatives of Corollary 5.3. The complete search over the 1,611 chosen base states finds no witness.

The five further deletions keep part of this symmetry. Deleting a vertex of Δ_1 from one of its orbits of size 11 leaves an automorphism group of order 5; Δ_{11} itself has an automorphism group of order 5, which fixes v_2, v_5 and v_{20}, so its three deletions $\Delta_{11} - v_2$, $\Delta_{11} - v_5$ and $\Delta_{11} - v_{20}$ keep it. In each of the five, this group maps a five-vertex base inducing the regular tournament on five vertices onto itself, so Corollary 5.5 again reduces the 8,031 base states to 1,611 chosen base states, and the complete search over them finds no witness. These five refutations are not formally certified; Appendix G.6 gives their cost.

The refutation of $\Delta_1 - v_{10}$ is also formally certified (Appendix B.1). The certificate uses the quotient of Corollary 5.5 but no anchoring, with a different base, $B' = \{y_1, y_3, y_4, y_5, y_9\}$, the five in-neighbours of y_0 in Y. H maps B' onto itself as well, and B' also induces the regular tournament on five vertices, so the certificate again keeps 1,611 of the 8,031 base states.

$\Delta_6 - v_5$ and $\Delta_{28} - v_5$ have no non-trivial automorphism, so none of Corollary 5.2, Corollary 5.3 and Corollary 5.5 applies and every base state has to be searched. Each proof is a complete search over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices, none of which extends to a witness. The refutation of $\Delta_6 - v_5$ is also formally certified, by the same pipeline as those of P_{19} and P_{23} (Appendix B.1).

All eight obstructions are *vertex-critical*: deleting any vertex of any of them leaves a 5-inducible tournament, so none yields a 21-vertex obstruction by a further deletion. For $\Delta_1 - v_{10}$ this takes only two deletions, one from each vertex orbit. Appendix C summarizes the wider search below 22 vertices, which has found none. Section 6.3 adds a ninth obstruction on 22 vertices, obtained from $\Delta_1 - v_{10}$ by reversing one arc.

6.3 Criticality across the Paley family

Arc-criticality implies vertex-criticality: reversing an arc incident to v leaves $T - v$ unchanged, and the restriction of a witness to $V \setminus \{v\}$ is a witness for $T - v$. Each cell of Table 5 therefore records only the *strongest* property established, at each of the two margins. "Not vertex-critical" also rules out arc-criticality.

Table 5: The strongest criticality property established for each Paley tournament, at each margin.

q	unit margin	unrestricted
19	**arc-critical**	5-inducible, so criticality does not apply
23	not vertex-critical	**arc-critical** (Section 6.6)
31	not vertex-critical	**not vertex-critical** (Section 6.5)
43	not vertex-critical	**not vertex-critical** (Section 6.5)

P_{19} is the only member critical at unit margin, and it is critical in the strongest sense: reversing any one arc makes it unit-margin inducible, which by the implication also gives vertex-criticality. At the other end, P_{31} and P_{43} fail to be vertex-critical at *both* margins. The interesting cell is $q = 23$: it is *vertex-critical unrestricted*, but *not* vertex-critical at unit margin. In other words, $P_{23} - v$ is 5-inducible, but not with unit margin, analogous to the following result.

The eight 22-vertex obstructions of Section 6.2 are vertex-critical unrestricted. Under its automorphism group, the 231 arcs of $\Delta_1 - v_{10}$ fall into five orbits: the arcs inside X, the arcs inside Y, the rungs $y_i \to x_i$, the other arcs from Y to X, and the arcs from X to Y. Reversing a rung yields a 5-inducible tournament, but reversing one of the other arcs from Y to X, $y_j \to x_i$ with $j - i \in \mathrm{QR}$, does not. The tournament it yields has no non-trivial automorphism, and a complete search over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices finds no witness. So $\Delta_1 - v_{10}$ is not arc-critical, and reversing one of its arcs gives a ninth obstruction on 22 vertices, not isomorphic to any of the eight of Section 6.2 or to the converse of any. This ninth obstruction is vertex-critical too. Deleting either endpoint of the reversed arc leaves $\Delta_1 - v_{10}$ minus a vertex, which is 5-inducible, and each of the other 20 deletions has an explicit witness. Whether reversing an arc of the remaining three orbits restores 5-inducibility is open, and so is the arc-criticality of the other seven obstructions of Section 6.2.

6.4 Paley(19) is not 5-inducible with unit margin

That P_{19} is 5-inducible is due to Bachmeier et al. (2019), who computed $\dim(Q_{19}) = 5$. Our search recovers it independently and contributes the negative half: it is *not* 5-inducible with unit margin. This is a complete refutation over all 2,200 base states of the base $\{1, 2, 3, 4, 6\}$, and it is the second statement we certify formally in Appendix B.1. These two facts supply the separation in Section 1: no arc of P_{19} can be unanimous, by the 3-cycle bound above, so it is 5-inducible with margin at most 3 and not with unit margin, and the inclusion of the unit-margin regime in margin at most 3 is strict.

6.5 Paley(31) and Paley(43) are not vertex-critical

Vertex-criticality is the natural minimality notion for an obstruction, and P_{43}, the first explicit non-5-inducible tournament of moderate order (Section 4), is the natural candidate.

Theorem 6.4. *Neither P_{31} nor P_{43} is vertex-critical for 5-inducibility.*

The orbit break. For any $q \equiv 3 \pmod 4$, $\mathrm{Aut}(P_q - v)$ is *exactly* the stabilizer of v in $\mathrm{Aut}(P_q)$. One inclusion is immediate; for the other, deleting v removes an out-arc from every vertex of $N^-(v)$ and from no vertex of $N^+(v)$, so $P_q - v$ has just two out-degrees, $(q-1)/2$ on $N^+(v)$ and $(q-1)/2 - 1$ on $N^-(v)$, and an automorphism preserves out-degree, hence those two sets, hence extends to P_q by fixing v. Translating so that $v = 0$, that stabilizer is $\{x \mapsto ax : a \in \mathrm{QR}\}$, cyclic of order $(q-1)/2$, and it splits the remaining $q - 1$ vertices into exactly two orbits of size $(q-1)/2$, namely QR and its complement. Two representatives, one from each, therefore meet every orbit, and the vertex anchoring of Corollary 5.3 applies with those two.

Each sweep runs over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices: $\{1, 2, 3, 4, 13\}$ in $P_{31} - v$ and $\{1, 2, 3, 4, 11\}$ in $P_{43} - v$. The two counts coincide because, as in Appendix G.2, the number of base states depends only on the induced subtournament and on the margin, and not on n.

Every arc of $P_q - v$ lies in at least $(q-3)/4$ directed triangles, so no arc of a majority witness could be unanimous, and margin ≤ 3 is not a restriction at all. The refutation at margin ≤ 3 is therefore an unrestricted refutation. We use the same approach for both $q = 31$ and $q = 43$.
We thus independently reconfirm Theorem 4.1, that P_{43} is not 5-inducible: if it were, so would be $P_{43} - v$, by deleting v from every voter's order.

6.6 Vertex and arc criticality for Paley(23)

Theorem 6.5. *P_{23} is arc-critical: reversing any single arc yields a 5-inducible tournament. Consequently it is vertex-critical.*

Let T^e denote P_{23} with the arc $(1, 2)$ reversed; since $\mathrm{Aut}(P_{23})$ is transitive on arcs, any two single-arc reversals of P_{23} are isomorphic, so we may pick $(1, 2)$ without loss of generality. The computation is a scan of the 8,031 base states of T^e, but now T^e is rigid, so Corollary 5.2 and Corollary 5.3 do not apply. The witness we found was verified independently of the search that produced it, by two checks on the five orders: each is a

permutation of the 23 vertices, and the majority tournament they induce agrees with T^e in every one of the 253 arcs and therefore differs from P_{23} in exactly e.

Vertex- and arc-criticality come apart, as far as they can. Arc-criticality implies vertex-criticality, so it is natural to ask whether the two coincide. They do not. Among the 110 vertex-transitive tournaments on 21 vertices, four are unit-margin obstructions, and all four are vertex-critical. However, arc-criticality takes all three of its possible values among those same four: two of them have every arc critical, one has exactly half, and one has none at all, so no statement of the form "this tournament is or is not arc-critical" captures the situation (full details in Appendix I).

6.7 Other structured tournament families

Beyond the Paley family we swept the structured families catalogued by McKay (2026) exhaustively for $n < 23$. Every verdict is negative for the bound, so we only report them briefly.

First, *the vertex-transitive family contains no majority obstruction at all for* $n < 23$: every one is 5-inducible, all at unit margin except P_{19} and four of the 110 on 21 vertices, which have majority witnesses. Since only a majority refutation can move $N(5)$, that family is closed as a source of candidates below 23, and the closure survives one-arc perturbation, since reversing any single arc of the four unit-margin obstructions, one representative per arc orbit, again leaves a majority-inducible tournament in all 40 cases.

Second, *both* doubly regular tournaments on 19 vertices, P_{19} and the one other of that order, are unit-margin obstructions and both are majority-inducible, so the separation that makes P_{19} interesting is not a property of P_{19} in particular. Nor is its criticality: the second one is *arc-critical* at unit margin too. Its automorphism group has order 3, so its 171 arcs fall into 57 orbits of size 3, and reversing the representative of any one of them yields a tournament with a unit-margin witness: 57 orbits and 57 witnesses. Arc-criticality at unit margin is therefore shared by both doubly regular tournaments of order 19.

Third, every regular tournament on at most 15 vertices is inducible at unit margin, over all $R_{15} \approx 1.8 \times 10^{10}$ of those on 15 vertices, and with the instance counts summing to OEIS A096368(7) (OEIS Foundation Inc. 2026c). Likewise, every self-converse tournament on 13 vertices is inducible at unit margin, over all $S_{13} \approx 9.5 \times 10^7$ of them. With the order-12 analysis of Appendix A, which covers every smaller order by inheritance, that settles every self-converse tournament on at most 13 vertices.

Fourth, the order 23 is complete for both families. Every vertex-transitive tournament on 23 vertices other than P_{23} is 5-inducible, while none of the 37 doubly regular ones is, at unit margin or unrestricted. Up to converse the 37 form 19 classes, and each was settled by a complete search (Appendix G.6).

Taken together with the Paley results above and the families of Appendix C, these searches turn up 75 tournaments, up to isomorphism and converse, that five voters can produce but not with unit margin: 2 doubly regular ones at order 19; at order 21, 3 vertex-transitive ones (the four of Section 6.6, two of which are converses of each other), 15 obtained from two of them by reversing one arc (Appendix I) and 42 near-doubly regular ones (Appendix C); at order 22, $P_{23} - v$; and at order 23, 11 circulant ones and P_{23} with one arc reversed. Several of them were found as a by-product of the unit-margin filter described at the start of this section.

One family needs no search. A tournament is *locally transitive* if every in- and out-neighbourhood induces a transitive subtournament. Every such tournament is k-inducible at unit margin for every odd $k \geq 3$, by an explicit construction rather than a computation; Appendix J provides it. This is the one structural family for which the inducibility question is settled outright. In Section 7 we discuss where the method may transfer and which questions remain open.

7. Discussion and open problems

In this section we discuss where the search method may apply beyond this problem (Section 7.1) and state the questions that our results leave open (Section 7.2).

7.1 Other applications of incremental depth-first search

Subject to the caveats of Appendix G, the engine's shape may transfer: a placement search in which each decision *refines*, rather than recomputes, the candidate sets of all remaining objects, with a dynamic minimum-remaining-values (MRV) order and an externally discharged symmetry break. The conditions are specific: the extensions of a partial solution must factor as a product that shrinks with each decision, and the symmetry group must be computable; otherwise a general-purpose solver may win.

Neither device is new in isolation, and the closest antecedent is Knuth's Algorithm X with dancing links (Knuth 2000). Its usual column-choice heuristic (the column with the fewest remaining ones, which Knuth credits to Golomb and Baumert) is our MRV order, and its link surgery does what our refinement does: the state is altered in place and put back on backtrack, so a node costs what changed rather than what the whole state contains. The constraint is what differs. Exact cover partitions, whereas an arc of a tournament imposes a threshold, at least three of five voters, so the search here is not an exact cover instance and dancing links does not apply to it as it stands.

Two design lessons do seem to generalize. First, the incremental refinement, not the heuristic order, carried most of the speedup here, a factor of 110 against a factor of a few. Second, a symmetry break should be *parameterized* by its representatives rather than hard-coded, so that its correctness obligation becomes a separate computation instead of an assumption buried in the search; Section 6.5 is a case where discharging that obligation required an explicit orbit calculation that the engine could not have performed for itself.

7.2 Conjectures and open problems

Question 1. *Is $N(5) = 22$, that is, is every tournament on 21 vertices 5-inducible?*

The first version of this paper (arXiv:2609.13924v1), whose upper bound was $N(5) \le 23$, conjectured that $N(5) = 23$. The obstructions on 22 vertices of Section 6.2 refute that conjecture, so we pose the question one order lower instead.

Both ends of $13 \le N(5) \le 22$ are constructive, unlike the counting bound of Section 3. The unit-margin obstructions at $n = 19$ and 21 cannot move the upper one, since every one of them is majority-inducible. The evidence below 22 comes mostly from the descendants of the doubly regular tournaments on 23 vertices (Appendix C): every tournament on 20 vertices obtained from one of them by deleting three vertices is 5-inducible, and so is every one on 21 vertices below any of the nine 22-vertex obstructions.

Every class we searched below 22 vertices was chosen because symmetry, a construction, or descent from a known obstruction made it searchable, and Appendix C lists them all. These classes cluster around the doubly regular tournaments, where every obstruction we know of lies. What they show is that no obstruction on 21 vertices lies *among them*; they say little about the roughly 3×10^{43} tournaments on 21 vertices.

Conjecture 2. *P_q is not 5-inducible for any eligible prime $q \ge 23$, i.e. $q \equiv 3 \pmod 4$.*

It holds at $q = 23$, 31 and 43, and only finitely many q are in doubt: $\alpha^*(P_q) = \mathrm{MAS}(P_q)/C$ tends to $1/2$, because Paley tournaments are quasi-random in the sense of Chung and Graham (1991, Section 6), and one of their equivalent quasi-random properties (P_9) is that every linear order of the vertices agrees with $(1/2 + o(1))\binom{q}{2}$ arcs. Hence the predictability eventually falls below $3/5$, past which the necessary condition $\alpha^* \ge 3/5$ refutes 5-inducibility with no search at all. What is missing is a bound on where that range ends; the values in Table 2 fall too irregularly to extrapolate one.

Open problem 1 (vertex- and arc-criticality). Whether P_q minus *two* vertices is 5-inducible, equivalently whether P_q minus one vertex is vertex-critical, is open at $q = 31$ and $q = 43$. For $q = 43$, $P_{43} - 2v$ is unique up to isomorphism, since $\mathrm{Aut}(P_{43})$ acts transitively on unordered pairs of vertices, and the screen of Section 4.2 extends to it. The two deleted vertices can be chosen after the screen rather than before: a pair of orders survives the deletion exactly when some set of at most two vertices meets every cyclic triangle that both orders double-back, a property invariant under Aut, so the reduction by a factor of 903 survives the deletion. The condition on the two voters that agree with the most arcs sharpens to $\delta_1 + 4\delta_2 \le 5\,\mathrm{MAS} - 3C$, where δ_1, δ_2 are their levels; the right-hand side is 45 for $P_{43} - 2v$.

Open problem 2 (the margin hierarchy). The first of unit margin $\subseteq$ margin $\leq 3 \subseteq$ unrestricted is strict (Sections 6.4 and 6.7); whether the second ever is, that is, whether some tournament is inducible by five voters but not at margin ≤ 3, is open. Lemma 6.1 forces such a tournament to have an arc in *no* directed triangle, which rules out every regular tournament. Conjecture 7.1 of our companion paper states that the inclusion is strict at every odd $k > 3$.

Conjecture 3. *For every odd k, every k-inducible tournament is $(k+2)$-inducible with unit margin.*

Conjecture 3 extends Conjecture 7.1 of our companion paper, which concerns the classes for a single k, to classes across different k. Adding an order together with its reverse raises every support by one, so k-inducibility with margin at most M implies $(k+2)$-inducibility with margin at most M. Conjecture 3 strengthens this to unit margin, and with the inclusions within each k it would nest every class at a lower k inside every class at a higher k, whatever the margins. It trivially holds at $k = 1$, since a transitive tournament is the majority of an order taken twice and its reverse once, with every arc carried two against one. At $k = 3$, Appendix A shows that every tournament on at most 12 vertices is 5-inducible with unit margin, so a counterexample needs at least 13 vertices. At $k = 5$, both doubly regular tournaments on 19 vertices, which are 5-inducible but not with unit margin, are 7-inducible with unit margin. In Section 8 we describe how every result in this paper can be reproduced.

8. Reproducibility

In this section we describe the public repository behind every result (Section 8.1) and say which parts of a rerun must match ours exactly and which may differ (Section 8.2).

8.1 The repository

The code, data and verdict files are at https://github.com/Leonardini/TournamentsBeyond5Voters. Its `CLAIMS.md` carries one row per statement made here, naming the artifact that holds the verdict and the command that re-derives it, and it ends with an explicit list of what the package does not establish. A permanent archive of the commit corresponding to the accepted version will be deposited on acceptance and its DOI recorded here.

The repository contains the search engine as a single C translation unit, the SAT encoding and cube tooling in Python, the certification driver, the independent witness verifiers, the tournaments as bit strings, and a verdict record for every result reported here. Each verdict records the exact command line, the base and break used, the coverage audit (the check that the cubes miss no base state, Appendix B.2), the runtime, and the two root hashes for the formally certified results. Three further folders hold the counting bounds (`counting_bounds/`), the proof for P_{43}, which rebuilds from one script (`paley43/`), and every check that the tournaments on at most 12 vertices are 5-inducible with unit margin (`n12_unit_margin/`). The last ships explicit profiles for the non-3-inducible tournaments on 8 and 9 vertices; at larger orders the same code regenerates them, for a single tournament in milliseconds and for all tournaments of an order in minutes at $n = 10$, in about 44 core-hours at $n = 11$ and in 2,616 cluster core-hours at $n = 12$.

8.2 What must and what may not replicate

The certified computations publish two root hashes, and only one of them is portable.

ROOT (CNF) commits to the SHA-256 of every cube's CNF, in cube order. These files are generated by the repository's own code from the tournament and the base, and are reproducible on any machine: a third party who regenerates the cube set *must* obtain this exact value. It is the meaningful cross-check, because it proves they solved the same problems.

ROOT (proofs) additionally commits to the LRAT proof bytes. These are *not* portable, as they depend on the exact SAT solver and version used. A mismatch here is not evidence of an error.

We have run that check: on a compute cluster, under a different compiler and CaDiCaL build, both refutations gave matching roots at 70.9 and 567.6 cluster core-hours against the 25.1 and 236.0 of Appendix G.6, and both coverage instances rebuilt to their committed CNF hashes and were refuted again.

Search runtimes: nodes replicate, hours do not. A three-way version of that distinction applies to the uncertified sweeps. *Verdicts* replicate under any correct implementation, which is the claim the paper makes. *Node counts* replicate exactly, but only with five choices held fixed:

1. the same tie-breaking rule in the variable order and in the domain refinement; ties are common and a different rule explores a different tree of the same size class but not the same size;
2. the same vertex numbering of the tournament;
3. the same base size and base;
4. the same resulting set of base states, which follows from (2) and (3);
5. the same margin regime.

The engine prints all of these except (2) in a header on every run (margin, variable order, tie-break mode, symmetry break, base size, base, base arc pattern and base-state count), so a run log is a self-describing fingerprint. Condition (2) is the tournament file itself, which is why the repository stores tournaments rather than regenerating them from their constructions. *Core-hours* replicate only in order of magnitude; Appendix G.5 measures a factor of 2 to 5 per core between the two machines we used.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Leonid Chindelevitch: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing. **Ararat Harutyunyan:** Conceptualization, Formal analysis, Writing – review and editing.

Data availability

The code, data and verdict files supporting the findings of this study are at https://github.com/Leonardini/TournamentsBeyond5Voters, described in Section 8.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) for exploratory reasoning, implementation assistance, and drafting and editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. The order-12 case analysis

Every tournament on 12 vertices is 5-inducible, so $N(5) \geq 13$. This appendix describes that computation, which starts from a census of order 11. It is one of the results in this paper that was not run on the laptop of Section 6: it ran on a shared cluster, and its runtime is converted as Appendix G.6 describes.

A.1 Every tournament on eleven vertices is 5-inducible

Our first census shows that every tournament on at most 11 vertices is 5-inducible, so $N(5) \geq 12$. Through $n = 10$ each tournament is decided by the integer program of Appendix G.1. At $n = 11$ no integer program was needed: of the $D_{11} = 903{,}753{,}248$ tournaments (OEIS Foundation Inc. 2026a), 382,625,248 are 3-inducible, and every one of the remaining 521,128,000 admits a *unit-margin* five-voter profile, found by a constraint search and re-checked by an independent verifier. A tournament with a non-3-inducible subtournament is itself non-3-inducible, and restricting a profile to a subset of the vertices preserves every pairwise split, so every non-3-inducible tournament on at most 11 vertices is 5-inducible with unit margin.

A.2 Reduction to one-vertex extensions

A direct census is out of reach: $D_{12} \approx 1.5 \times 10^{11}$ classes, 170 times the order-11 census of Appendix A.1. We use that census as the outer loop instead. Let us write $L + S$ for the tournament obtained from an order-11 tournament L by adding a vertex v whose out-neighbourhood is $S \subseteq V(L)$. Two observations simplify the analysis.

(i) Every order-12 tournament is $L + S$ for some order-11 L and some S.

(ii) Restricting a profile that realizes $L + S$ to $V(L)$ gives a profile that realizes L.

By (i), enumerating all D_{11} order-11 classes and settling all 2,048 extensions of each settles every order-12 tournament; all of the former are 5-inducible by the census of Appendix A.1, which is what makes the outer loop available. By (ii), a profile realizing $L + S$ may be sought by *inserting* v into a profile realizing L with no loss of generality, since every profile for $L + S$ arises that way. That is what phase 1 uses.

A.3 Three phases

The census runs in three phases, fastest first.

Phase 1, the covering search. Rather than 2,048 separate instances per order-11 tournament, one search enumerates profiles realizing L *at unit margin* and, for each, records every out-neighbourhood obtainable by inserting v into its five orders: v's position in each order is a threshold, and the support of $v \to u$ is the number of orders that place v above u, so one traversal certifies many extensions at once. This is the engine of Appendix G.3 with v's placement read off rather than fixed in advance. The margin makes this phase pay

twice: it keeps the enumeration of profiles for L small, and a unit-margin witness is in particular a majority witness, so every extension the phase covers is settled in both regimes at once. The phase is one-sided: an extension it covers is certified 5-inducible, but one left over is not refuted. This phase settles every one of the 2,048 extensions of 99.5% of the order-11 tournaments.

Phase 2, deciding what is left. Each extension phase 1 leaves open is materialized as an order-12 tournament $L + S$ and decided by the same engine, again at unit margin, where a witness settles it positively. About one extension in 200,000 reached this phase, and every one of them is 5-inducible with unit margin.

Phase 3, plain majority. Only a *majority* refutation would be a counterexample, so any instance negative at unit margin is re-run without the margin restriction. Phase 3 was never invoked: every tournament was settled at unit margin, so the weaker question never had to be asked.

A.4 One tournament of each converse pair

T is 5-inducible if and only if its converse $\bar{T}$ is (Section 1). Moreover $\overline{L+S} = \bar{L} + (V(L) \setminus S)$, so settling all 2,048 extensions of L settles all 2,048 of $\bar{L}$ as well. We therefore keep one tournament of each converse pair, by the rule that L is kept if and only if $\mathrm{canon}(L) \leq \mathrm{canon}(\bar{L})$. This keeps exactly one of each pair together with every self-converse tournament, $(D_{11} + S_{11})/2$ in all, where S_{11} is the number of self-converse order-11 classes; this saves a factor of $\lessapprox 2$. We also check the number of tournaments kept is $(D_{11} + S_{11})/2$.

Total runtime: 2,615.6 core-hours. The search code, the per-part records and the roll-up are at `n12_unit_margin/n12_cover` in the repository of Section 8.

Appendix B. The formal certificates

In this appendix we describe how the four certified negative results are proved (Appendix B.1) and what the certificates do and do not establish (Appendix B.2).

B.1 The certified refutations

All four machine-checked results follow an established design. *Cube-and-conquer* is due to Heule, Kullmann, Wieringa and Biere (2012): split a hard instance into many partial assignments (*cubes*), hand each to a CDCL solver, and let the splitting supply the global structure the solver cannot find for itself. The standard of *proof* is the one Heule, Kullmann and Marek set for the Boolean Pythagorean triples problem (Heule et al. 2016), later re-checked with a formally verified checker (Cruz-Filipe et al. 2019), and Schur number five (Heule 2018) and the Keller conjecture (Brakensiek et al. 2020) after it: every subproblem emits a machine-checkable refutation, validated by an independent checker rather than trusted. We claim no methodological novelty.

What an LRAT proof is. A solver's verdict of unsatisfiability is not self-evidently checkable, so the solver writes a *proof log*: a file recording, in order, the clauses it derives, each of which must follow from the input formula together with the clauses already recorded, the last being the empty clause. A separate checker replays that file and confirms every step, so one trusts the checker, not the solver. The available formats record essentially this same derivation and differ mainly in the cost of the replay. We use LRAT (Cruz-Filipe et al. 2017), the format CaDiCaL emits natively and `lrat-trim` consumes, in which each derived clause lists the earlier clauses it follows from, so checking it needs no search (Pollitt et al. 2023).

Two differences are consequences of our setting. Our cubes are *base states*, a combinatorial decomposition, which lets the coverage of the cube set be certified separately (Appendix B.2) instead of resting on a splitting heuristic. And where Heule et al. (2016) published a cube certificate from which its refutations can be regenerated, we *verify and discard*: each LRAT proof is checked, hashed and deleted, and we only publish the hash value. The choice suits the scale. Verifying the Boolean Pythagorean triples proof costs about 13,000 core-hours to regenerate the refutations from the published cube certificate and about 16,000 to check them (Heule et al. 2016). Our certificates are orders of magnitude smaller: 25.1 core-hours for P_{19}, 236.0 for P_{23}, 1,292.7 for $\Delta_6 - v_5$ and about 732 for $\Delta_1 - v_{10}$ cover solving *and* checking. Anyone replicating the computation can compare the portable root values (Section 8.2).

For P_{19} at unit margin the instance is the dissent-Boolean encoding of Appendix G.1 at $q = 19$, $k = 5$: 855 variables and 12,255 clauses. The cubes are the 142,251 base states of the 6-vertex base $\{1, 2, 3, 4, 5, 12\}$, of which 22,876 survive the arc-orbit break, and each is handed whole to CaDiCaL 2.0.0 with `--lrat=true --checkproof=0`. Each LRAT proof is checked by `lrat-trim`. The runtime was 2.64 hours on ten cores. Its coverage half is the same construction as the one below, at 143,032 clauses, and solves in 7.56 seconds.

For the P_{23} refutation of Section 6.1, the base is $\{1, 2, 3, 4, 6, 7\}$ with 3,414,729 base states, of which 343,896 survive the arc-orbit and non-arc-orbit breaks; every one is unsatisfiable, with an LRAT proof independently verified by `lrat-trim`. The runtime was 21.72 hours on eleven cores. The coverage half of the proof is a single unsatisfiable instance of 350 constraint clauses, 356 symmetry-breaking clauses and 3,414,729 negated cubes, one per base state, since coverage must see *every* base state whereas the orbit breaks apply to the search half only. It solves in just under 20 minutes on one core and emits a 2.05 GiB LRAT proof.

For the $\Delta_6 - v_5$ refutation of Section 6.2 the instance is the dissent-Boolean encoding at $n = 22$ and plain majority: 1,155 variables and 17,710 clauses. The tournament has no non-trivial automorphism, so there is no orbit break, and all 742,374 base states of the six-vertex base $\{1, 2, 3, 4, 13, 19\}$ (in the labelling of Δ_6) are cubes; every one is unsatisfiable, with an LRAT proof independently verified by `lrat-trim`. The runtime was about 118 hours on eleven cores. The coverage half is the same construction as for P_{23}, with 742,374 negated cubes, and solves in 82.8 seconds on one core. This refutation is the most expensive of the four, although its tournament is smaller than P_{23}: with no automorphisms there is nothing to break, so every base state reaches the solver, against one in ten for P_{23}.

For the $\Delta_1 - v_{10}$ refutation of Section 6.2 the instance is the same encoding as for $\Delta_6 - v_5$, at $n = 22$ and plain majority. The cubes are the 1,611 orbit representatives under H of the 8,031 base states of the five-vertex base $B' = \{13, 15, 16, 20, 22\}$ (in the labelling of Δ_1), one per orbit, as Corollary 5.5 licenses; every one is unsatisfiable, with an LRAT proof independently verified by `lrat-trim`. The cubes were solved in groups of five, part on the laptop and part on the cluster, at about 732 core-hours (Appendix G.6). The coverage half is the same construction as for P_{23}, with 8,031 negated cubes among its 8,467 clauses over 90 variables, and solves in 0.08 seconds on one core. A cube here fixes the voters' orders on five vertices rather than six, so the cubes are about 460 times fewer than for $\Delta_6 - v_5$, and each is correspondingly harder.

B.2 What the certificates do and do not establish

The trust chain has four links:

Table B.1: The four links of the trust chain.

link	status
every live cube is unsatisfiable	machine-checked: CaDiCaL, then `lrat-trim`
the cubes cover every lex-canonical base assignment	machine-checked: LRAT, verified by `lrat-trim`
(L1) orbit anchoring loses no generality	human-checked: Corollary 5.2
(L2) lex-ordering the voters loses no generality	human-checked: Corollary 5.4
(L3) one base state per orbit loses no generality	human-checked: Corollary 5.5

The human-checked links are proved in Section 5.1. Each certificate uses only some of them: P_{19} and P_{23} use (L1) and (L2), $\Delta_1 - v_{10}$ uses (L2) and (L3), and $\Delta_6 - v_5$ uses (L2) alone. Corollary 5.2 is what licenses solving only the *live* cubes, in its arc/non-arc form: $\mathrm{Aut}(P_q)$ is transitive on arcs and on non-arcs, so a voter's top pair may be assumed to be a fixed arc or a fixed non-arc. Corollary 5.4 licenses the lex chain in the coverage computation. Corollary 5.5 licenses solving one base state per orbit of H for $\Delta_1 - v_{10}$, while the coverage computation still runs over all 8,031 base states. $\Delta_6 - v_5$ has no automorphism to exploit, so neither orbit argument is used. These results are human-checked links because nothing in the pipeline checks them; the depth-first engine inherits them, but shares no implementation with this one, so their agreement is evidence about the implementations, not the results.

Only some results carry a certificate. Table B.2 says, for each result, what decided it and what independent check, if any, stands behind the verdict. A negative verdict from the depth-first engine alone rests on the

engine's correctness; a positive one never does, because every witness is a profile that a separate program checks arc by arc.

Table B.2: What each result rests on. The first six rows are refutations (not 5-inducible, at unit margin where stated); "engine" is the depth-first search of Appendix G, run twice on P_{23} with different bases and anchors, and a verifier rebuilds the majority tournament of a profile and compares it with the target, sharing no code with the search.

result	decided by	independent check
P_{23}	engine, twice	LRAT certificate
$\Delta_1 - v_{10}$, $\Delta_6 - v_5$	engine	LRAT certificate
P_{19}, at unit margin	engine	LRAT certificate
the other 22-vertex obstructions	engine	none
P_{27}, P_{31}, $P_{31} - v$	engine	none
P_{43}	proof, two programs	$P_{43} - v$, by the engine
all orders ≤ 11 inducible	ILP, constraint search	verifier, per profile
all of order 12 inducible	extension search	check inside the search
positive results of Section 6	engine	verifier, per witness
counting bounds	proof	none needed

Appendix C. The search below 22 vertices

This appendix summarizes the search for an obstruction on fewer than 22 vertices, which would improve the upper bound of Section 6.2. It has found none.

The search runs downward, because 5-inducibility is inherited by subtournaments: restricting a witness for T to a subset of the vertices gives a witness for the subtournament. An obstruction on fewer vertices therefore lies below some larger one, and the natural place to look is among the vertex deletions of known obstructions. A tournament is settled positively as soon as one base state yields a witness, which is cheap, and negatively only by a complete search, which is not; so the tournaments of each order are first screened for witnesses, and only tournaments that survive the screen are searched further.

The starting points are the 36 doubly regular tournaments on 23 vertices other than P_{23}, all of which are obstructions (Section 6.7). P_{23} itself is excluded, since $P_{23} - v$ is 5-inducible (Section 6.6) and so, by inheritance, is every tournament below it. Up to isomorphism and converse, the tournaments below the 36 are as follows.

Table C.1: The tournaments below the 36 doubly regular tournaments on 23 vertices, up to isomorphism and converse.

vertices	deleted	classes	status
22	1	378	8 searched completely, all obstructions (Section 6.2)
21	2	4,108	1,192 settled, all 5-inducible
20	3	28,724	all 5-inducible

At 22 vertices all 378 classes were first screened at up to 50 base states each, without a single witness, and the eight complete searches were chosen from among them. At 21 vertices the 1,192 settled classes include all 506 two-vertex deletions of Δ_6 and Δ_{28}, so every 21-vertex subtournament of either parent is 5-inducible, and in particular $\Delta_6 - v_5$ and $\Delta_{28} - v_5$ are vertex-critical. The other settled classes are a stratified random sample of 400; all 53 classes below Δ_{11}; every class obtained by deleting v_5 together with one other vertex, the deletion that produced those two obstructions; and 19 classes settled by inserting the missing vertex into a known witness of one of their 20-vertex subtournaments, as in phase 1 of Appendix A.3; the remaining 2,916 classes are open. At 20 vertices every class is 5-inducible, and so, by inheritance, is every tournament on 19 vertices or fewer below the 36.

Two obstructions on 25 vertices from other families were descended the same way: P_{27} with two vertices deleted, and the circulant tournament on 25 vertices in which every arc lies in six directed triangles. Neither is 5-inducible, by complete searches, and every 21-vertex subtournament of either is 5-inducible, so neither family yields an obstruction below 22.

Other families on 21 vertices. We also searched families on 21 vertices that a symmetry or a construction makes enumerable, all of them close to doubly regular. A regular tournament on 21 vertices cannot be doubly regular, since $(21-3)/4$ is not an integer; we call it *near-doubly regular* if every pair of vertices has 4 or 5 common out-neighbours. The table below lists every family we searched on 21 vertices, again up to isomorphism and converse, with the number of classes settled; the vertex-transitive tournaments on 21 vertices, a family enumerated in full, are covered in Section 6.7. Every settled class is 5-inducible, by a witness verified independently of the search; none was settled negatively.

Table C.2: The families searched on 21 vertices, up to isomorphism and converse.

family on 21 vertices	classes	settled
two vertices deleted from a doubly regular tournament on 23 vertices	4,108	1,192
four vertices deleted from the circulant tournament on 25 vertices	286	286
six vertices deleted from P_{27}	150	150
near-doubly regular, rigid, drawn at random	78	78
near-doubly regular, with a bisector and an automorphism of order 5	28	28
near-doubly regular, with a fixed-point-free automorphism of order 3	31	31

The 31 classes of the last family are likely, but not confirmed, to be all of it: the enumeration found no new class after its first 225 solutions, and we stopped it at 4,567 without running it to completion. Two of the 31 are bisectable and the other 29 are not. In the two families with an automorphism, several classes are not 5-inducible at unit margin and need a majority witness: 12 of the 28 with a bisector, and 30 of the 31 with an automorphism of order 3.

Appendix D. Exact enumeration counts

In this table, D_n, R_n and S_n are the numbers of isomorphism classes of all, of regular, and of self-converse tournaments on n vertices. They match OEIS A000568 (OEIS Foundation Inc. 2026a), A096368 (OEIS Foundation Inc. 2026c), and A002785 (OEIS Foundation Inc. 2026b).

Table D.1: Numbers of isomorphism classes of all, regular and self-converse tournaments.

n	D_n	R_n	S_n
11	903,753,248	1,223	279,968
12	154,108,311,168	—	1,492,288
13	—	1,495,297	95,458,560
15	—	18,400,989,629	—

Appendix E. The MAS algorithm: orbit subset-DP, meet-in-the-middle, and the negation/converse join key

This appendix describes the dynamic program that certifies $\mathrm{MAS}(P_{43}) = 543$ (Section 4.1) and enumerates the level-≤ 1 shell (Section 4.2).

E.1 Base recurrence (Held–Karp 1962). For $S \subseteq V$ let $g(S) = \mathrm{MAS}(T[S])$. The last vertex v of an order of S gains its in-arcs from $S \setminus \{v\}$, so

$$g(\varnothing) = 0, \qquad g(S) = \max_{v \in S} \big[\, g(S \setminus \{v\}) + |\{u \in S \setminus \{v\} : u \to v\}| \,\big],$$

and $\mathrm{MAS}(T) = g(V)$.

E.2 Automorphism quotient. As $T[\sigma S] \cong T[S]$ for $\sigma \in \mathrm{Aut} = \{x \mapsto ax + b : a \in \mathrm{QR}\}$, we keep one representative per orbit: 4.9×10^9 states up to $|S| = 21$ (almost 903 times fewer). The representative is the minimum bit-mask over the images that send a vertex of maximum out-degree within S to 0. Layer $j + 1$ comes from layer j: a stored S yields, for each $v \notin S$, the record $(\mathrm{canon}(S \cup \{v\}), g(S) + |\{u \in S : u \to v\}|)$, with the gain computed before canonicalizing; equal keys keep the largest value, which is $g(S \cup \{v\})$ by the recurrence.

E.3 Meet-in-the-middle at the equator. Storing every layer up to $|S| = 43$ is out of reach, so we build layers 0 to 22 only and split every order in the middle. Let π be an order of V and S the set of its first 21 vertices (the **equator** is $h = 21$). An arc of T lies inside S, inside $V \setminus S$, or between the two, and an arc between them is forward in π exactly when it goes from S to $V \setminus S$. Hence

$$\mathrm{fwd}(\pi) = \mathrm{fwd}(\pi|_S) + \mathrm{cross}(S) + \mathrm{fwd}(\pi|_{V \setminus S}),$$

where $\mathrm{cross}(S)$ counts the arcs from S to $V \setminus S$, and maximizing the two halves separately gives

$$\mathrm{MAS}(T) = \max_{|S|=21} \big[\, g(S) + \mathrm{cross}(S) + g(V \setminus S) \,\big].$$

The first term is read from layer 21. The suffix $V \setminus S$ has 22 vertices, but the tables were built from the front of an order, so we read the suffix through a twin order. Negation $x \mapsto -x$ maps T onto its converse, because -1 is a non-residue; so reversing π and negating every vertex gives an order π^* with $\mathrm{fwd}(\pi^*) = \mathrm{fwd}(\pi)$, whose first 22 vertices form the set $-(V \setminus S)$. In particular $g(V \setminus S) = g(-(V \setminus S))$, which layer 22 stores under the key $\mathrm{canon}(-(V \setminus S))$.

E.4 The join: the negation/complement commutes with the quotient by Aut**.** Layer 21 holds one representative per orbit, so the maximum above must be taken over orbits, and the key of the suffix must not depend on which representative was stored. Let $\nu(S) = -S$, $\kappa(S) = V \setminus S$ and $\varphi = \nu \circ \kappa$, so that the suffix of S is looked up at $\mathrm{canon}(\varphi(S))$. (i) κ commutes with every bijection of V; (ii) $\nu \circ \sigma_{a,b} = \sigma_{a,-b} \circ \nu$, so negation normalizes Aut while lying outside it; hence (iii) $\varphi \circ \sigma_{a,b} = \sigma_{a,-b} \circ \varphi$. So if S and S' lie in one orbit, so do $\varphi(S)$ and $\varphi(S')$: $\mathrm{canon}(\varphi(S))$ is constant on orbits, and φ induces a one-to-one map from the orbits of 21-sets to the orbits of 22-sets. As g and cross are constant on orbits as well, the maximum over all 21-sets equals the maximum over the stored representatives.

We stream layer 21 and form, for each representative S, the pair $(\mathrm{canon}(\varphi(S)), g(S) + \mathrm{cross}(S))$; we sort these pairs by key and merge them with the stored layer-22 table, which is sorted by the same key. Each match gives $\mu(S) = g(S) + \mathrm{cross}(S) + g(\varphi(S))$, the largest number of forward arcs of an order whose first 21 vertices form S, and $\mathrm{MAS}(T) = \max_S \mu(S) = 543$. Backtracking through the stored values from the splits with $\mu(S) \geq 542$ lists every order of level ≤ 1: this is the shell of Section 4.2, and its orbit count matches the census of Appendix F.

Appendix F. Counts for the Paley(43) proof

This appendix gives the counts behind the proof that $N(5) \leq 43$ (Section 4): the structural counts of Paley(43) and the certified value MAS = 543 (Table F.1), then the sizes of the shell and of the screen.

The level-≤ 1 shell has 1,841,303 Aut-orbits and 1,662,696,609 orders: 19,651 orbits (17,744,853 orders) at level 0 and 1,821,652 orbits (1,644,951,756 orders) at level 1. In the screen (Section 4.2), the shell orders have 4,709,640 distinct restrictions to the cyclic triangles inside the window $\mathcal{W}$, and 5,092,111 pairs of these restrictions are disjoint. Only the 678,686 restrictions that belong to such a pair, and the 347,694,990 shell orders that have one of them, need the exact check. It compared 4,376,325,129 pairs of an orbit representative and such an order, and found no pair with disjoint double-back sets (0 of 4,376,325,129).

Appendix G. The search engine

Integer programming and satisfiability are both central to this project. Every optimum reported in the companion paper is certified by an integer or linear program, and the integer program of Appendix G.1

Table F.1: Structural counts and certified quantities for Paley(43).

quantity	formula	value
arcs C	$q(q-1)/2$	903
cyclic triangles N_Δ	$(q^3-q)/24$	3311
$\lvert\mathrm{Aut}(P)\rvert$	$q(q-1)/2$	903
out-degree	$(q-1)/2$	21
triangles per arc	$(q+1)/4$	11
$\mathrm{MAS}(P)$	dynamic program (Appendix E), certified	543
$\alpha^* = \mathrm{MAS}/C$		181/301
slack $5\,\mathrm{MAS} - 3C$		6
forced top-two level	$\lfloor \mathrm{slack}/4 \rfloor$	1

below is that paper's k-inducibility formulation with antisymmetry substituted out; the only machine-checked theorem in this paper is a SAT computation from end to end, with CaDiCaL (Biere et al. 2024; Pollitt et al. 2023) emitting LRAT proofs that `lrat-trim` (Pollitt et al. 2023) checks. The claim here is narrow: on this one decision problem, at this one range of sizes, neither tool reaches the answer while a search built around the problem's own structure does, and the same SAT machinery is then indispensable for turning that answer into a proof.

G.1 Integer programming and SAT are both too slow

The natural integer program has one binary $x_{v,ij}$ per voter v and pair $i < j$, meaning that v ranks i above j, with $x_{v,ji}$ read as $1 - x_{v,ij}$ so that antisymmetry is structural. Each voter is forced to be a linear order by forbidding both orientations of every 3-cycle,

$$0 \;\le\; x_{v,ij} + x_{v,jl} - x_{v,il} \;\le\; 1 \qquad (i < j < l),$$

and the profile is forced to induce T by one majority row per arc, $\sum_v x_{v,ij} \ge (k+1)/2$. This is a pure feasibility problem. At $q = 23$, $k = 5$ it has 1,265 binaries, 17,710 transitivity rows and 253 majority rows.

The obstruction is not a deficiency of the solver. Let us set $x_{v,ij} = 3/5$ on every arc of T and $2/5$ on every non-arc, the same for all voters. Every cyclic triple then reads $4/5$ and every transitive one $3/5$, both interior to $[0, 1]$, while every majority row is tight at exactly 3. The linear programming (LP) relaxation is therefore feasible, and since the problem carries no objective, its feasibility yields no bound whatever: branch-and-bound is left with nothing to prune by except infeasibility discovered deep in the tree. That fractional point is also invariant under both symmetry groups acting on the problem (the $k!$ voter relabellings and $\mathrm{Aut}(T)$), so it supplies no branching direction either, and orbital branching recovers only the $k!$ factor. CPLEX and Gurobi both failed to close instances at $n = 19$ within a day.

The satisfiability encoding is *dissent-Boolean*: one variable per voter v and arc $i \to j$ of T, true when v ranks j above i and so dissents from that arc. Each arc bounds its dissents, at most $(k-1)/2$ of them for majority and exactly $(k-1)/2$ for unit margin, and each voter's dissent set is forced to leave a linear order triple by triple: a tournament is transitive exactly when it has no directed 3-cycle, so it is enough to forbid, for every voter and every triple, the two dissent patterns that would make that triple cyclic. For $k = 5$ that is $5\binom{n}{2}$ variables, 15 clauses per arc at unit margin and 10 per triple. It is compact and propagates well locally, and it is what the certified refutations of Appendix B are built on.

Table 4 of Bachmeier et al. (2019) reports their `Sat-Check-k-Majority` on uniform random tournaments, tabulating only entries whose average stays under 20 seconds. The $k = 5$ column reads

Table G.1: Average time of Sat-Check-k-Majority at $k = 5$ on uniform random tournaments (Bachmeier et al. 2019, Table 4).

n	18	19	20	21	22	23	24
seconds	0.23	0.35	0.54	5.87	11.07	18.95	—

so on *random* instances the method leaves the table between $n = 23$ and $n = 24$, roughly doubling per added vertex. Random tournaments are the easy case: no automorphisms to rediscover and, being far from extremal, many witnesses. On the structured instance of the same order, P_{23}, the same solver did not terminate within a cumulative six weeks. The gap between 19 seconds and six weeks at equal n is the cost of symmetry.

A better encoding and an explicit symmetry break both help substantially but are not enough (results not shown). What does work is supplying the solver with the decomposition itself: cube-and-conquer over base states (Appendix G.2), which is how the refutations of Appendix B are certified. Otherwise a CDCL solver must discover it for itself, and cannot exploit $\mathrm{Aut}(T)$ at all without an explicit break in the encoding.

G.2 Base states

Every search starts by fixing how the voters rank a handful of vertices and then looks for ways to add the remaining ones. We choose a *base* $B \subseteq V$ of b vertices ($b = 5$ or 6), and write $T[B]$ for the sub-tournament induced on B. A *base state of* B (we omit B when the context makes it clear) is a k-voter profile over B that is consistent with $T[B]$, i.e. k orderings of B whose majorities reproduce $T[B]$ at the margin being asked for. Base states differing only by a relabelling of the voters are counted once (see Corollary 5.4).

Each base state of B defines a self-contained subproblem: the rankings of B are given, and the question is whether the remaining $n - |B|$ vertices can be inserted so that every arc of T has the right support, so the search splits into independent components across base states. How many there are depends only on $|B|$, on $T[B]$ and on the margin regime, but *not* on n; thus, with the five-vertex bases we use on the Paley tournaments, which induce regular sub-tournaments, there are 8,031 of them in both P_{23} and P_{47}.

G.3 Depth-first placement search

The search maintains a partial profile: a set $S \subseteq V$ of *placed* vertices together with, for each voter, a linear order of S. It begins from a base state, which places B, and extends by inserting one unplaced vertex at a time. Inserting v means choosing, for each voter independently, a slot among the $|S| + 1$ available; the *domain* $D(v \mid S)$ is the set of k-tuples of slots consistent with every arc between v and S. A partial profile is extendable only if every unplaced vertex has a non-empty domain, and domains shrink monotonically as S grows, reducing the search to a constraint-propagation problem. Two decisions in the pseudocode below make it efficient (see Appendix G.4 for more details). First, the search always inserts an unplaced vertex of currently smallest domain (the minimum-remaining-values, or MRV, rule), the domain sizes being recomputed at every node rather than fixed once per base state; second, a child's domains are refined from its parent's. The pseudocode also shows where the symmetry reductions of Section 5.1 enter (Appendix G.4).

G.4 What makes the search fast

Five choices make the search of Appendix G.3 fast, and the second is the one that made $n = 19$ and $n = 23$ reasonable to compute.

A dynamic variable order. At every node we insert the unplaced vertex minimizing $|D(v \mid S)|$, breaking ties toward the vertex whose arcs to S are most imbalanced. The important part is that the order is recomputed at every node rather than fixed per base state: on Paley instances a static order is not competitive, because how constrained a vertex is depends strongly on which part of the tournament has already been placed.

Incremental domain refinement. Recomputing $D(v \mid S)$ from scratch at each node is the dominant cost of a naive implementation; instead the child's domain is derived from the parent's by the three cases in `REFINE` above, after which the single new arc filters the lifted tuples by their support. Child domains are

```
ALGORITHM 1.  Placement search for k-inducibility of T

Input :  tournament T on V;  k voters;  margin (majority or unit);  base B
         Q: chosen base states of B    // sorted (Cor 5.4), one per H-orbit (Cor 5.5)
         R: anchoring representatives  // vertices (Cor 5.3), pairs (Cor 5.2), or none
Output:  a witness profile, or a refutation

MAIN
  for each base state in Q  (a lex-canonical assignment of the base B to k orders):
      S <- B
      D(v | S) <- fresh enumeration, for every v not in S
      if any D(v | S) is empty:  continue     // this base state is dead
      if DFS(S) succeeds:        return its witness
  return "not k-inducible"                    // no base state extends

DFS(S)
  if S = V:  return the profile               // all vertices placed
  if every vertex of R is placed and no voter is anchored:
      return failure                          // Cor 5.2/5.3; at the root if R is in B
  v* <- argmin  |D(v | S)|   over v not in S  // MRV, recomputed at EVERY node
        ties broken toward  max | |out(v) & S| - |in(v) & S| |
  for each slot tuple p in D(v* | S):
      insert v* at p in each of the k orders;  S' <- S u {v*}
      for each w not in S':
          D(w | S') <- REFINE( D(w | S), p, w, v* )
          if D(w | S') is empty:  undo;  continue with the next p
      if DFS(S') succeeds:  return its witness
      undo                                    // arena pointer reset
  return failure

REFINE(D, q, w, u)                            // q = the slot tuple just used for u
  D' <- {}
  for each p in D:
      lift p to p' voter by voter:
          p_i <  q_i  ->  p'_i = p_i        // w stays before u
          p_i >  q_i  ->  p'_i = p_i + 1    // w stays after u, shifted by the insertion
          p_i == q_i  ->  BOTH are legal, so the tuple splits in two
      keep those lifts whose support  c_w(u) = #{ i : w before u }
      meets the requirement of the single new arc between w and u
  return D'
```

therefore refinements of parent domains, computed in time proportional to the surviving tuples rather than to the whole slot space, and domains live in a stack-disciplined arena indexed by (depth, vertex) so that backtracking is a pointer reset. Like for like this is worth a factor of about 110 in wall time.

Symmetry breaking. We reduce the base states using the voter relabellings (Corollary 5.4) and the orbits of a subgroup of $\mathrm{Aut}(T)$ mapping the base onto itself (Corollary 5.5), and we restrict the search using the anchors (Corollary 5.2 or 5.3), tested at every node once all representatives are placed; Section 5.1 and Figure 2 give the details.

Early termination and caching by arc pattern. A quick per-voter agreement bound, obtained by counting the agreement between voter i and the arcs between v and S two ways, proves $D(v \mid S)$ empty in $O(k|S|)$ time without enumerating it. And the per-base-state work is keyed by the *arc pattern* of $T[B]$ rather than by B itself, so base-state counts, which depend only on that pattern, are reused across every copy of the same base class and across every q.

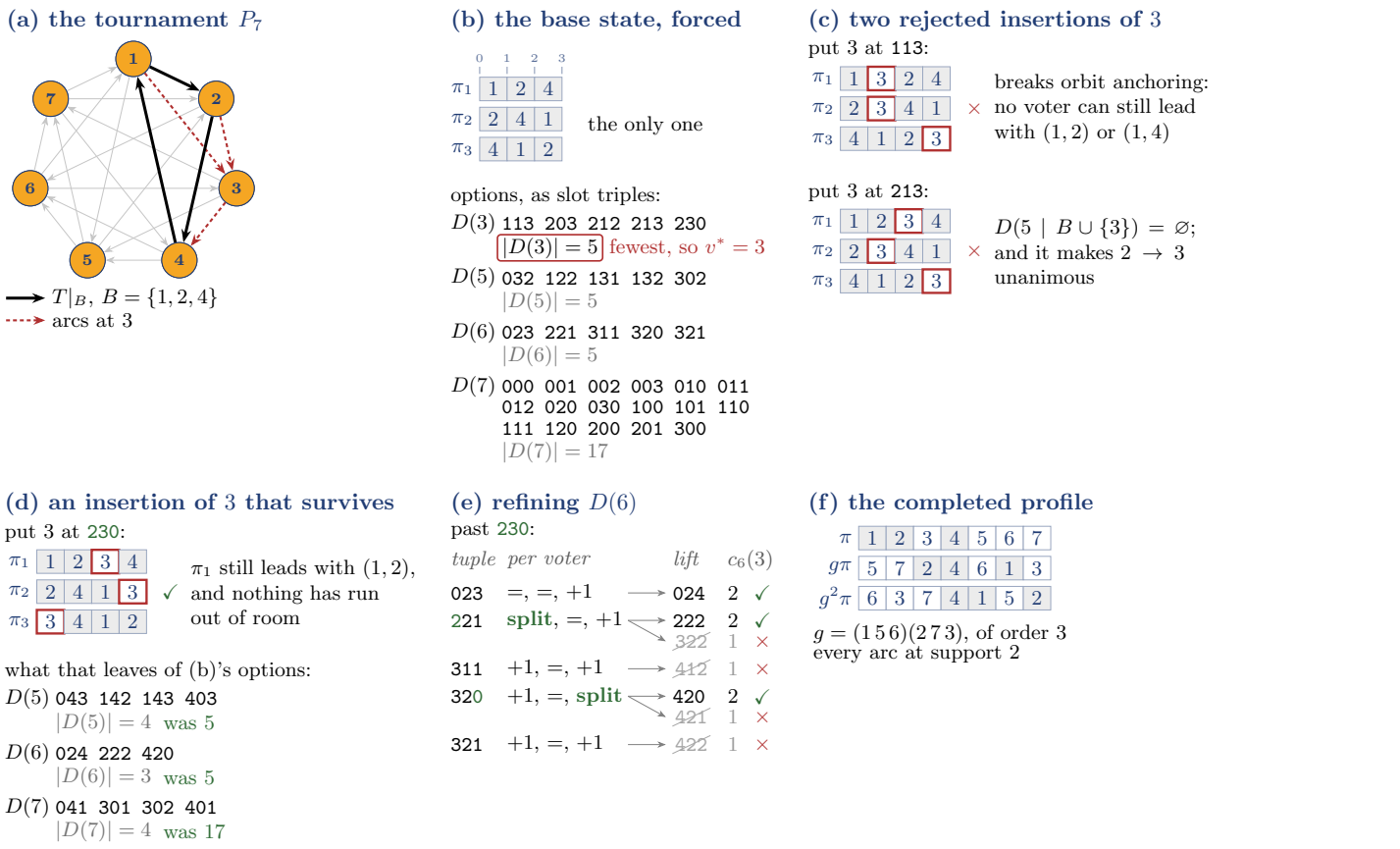


Figure G.1: Algorithm 1 on P_7 at $k = 3$, which is 3-inducible, so the run ends in a witness. A *slot triple* is one gap per voter, of the 64 available, and a vertex's domain is those triples its arcs to B admit. Here the base is $B = \mathrm{QR}$, the squares modulo 7, and Corollary 5.2 guarantees that at least one voter must rank 1 first and either 2 or 4 second. Ranking 4 second forces the other two voters to 241, the only order getting both $4 \to 1$ and $2 \to 4$ right, and $1 \to 2$ is then left with support 1; so the base state in (b) is the only one. **(a)** The base $T[B]$ on $B = \{1, 2, 4\}$ in bold; dashed, the arcs at the first vertex inserted. **(b)** Steps 2–4: the base state is forced, and MRV takes the smallest domain. **(c)** Two insertions the search rejects, one by Corollary 5.2 and one by leaving 5 no slot at all. **(d)** One that survives, and what it leaves of (b). **(e)** REFINE lifts each parent tuple past the slot just used (unchanged below it, shifted above it, split where the two coincide, which is what green marks), and the one new arc then keeps or kills each lift. **(f)** The witness, every arc at support exactly 2. Without Corollary 5.2 the search returns all 7 witnesses, a single orbit of $\mathrm{Aut}(P_7)$, of order 21, so the profile is unique up to symmetry, as Shepardson and Tovey (2009) report; π generates the other two under the g shown, the map $x \mapsto 2x + 4$ of $\mathbb{F}_7$.

Choosing the base. The base set is not a free parameter, and the fewest live base states is not the fastest choice: at $q = 31$ the class with 4,007 live states beat the class with 2,537, whose states are individually slower, so we rank the twelve 5-vertex classes by the product of the live count and a sampled time per state.

G.5 How the runtime scales in q

A note on $q = 27$. The series below includes $q = 27$, the one prime power among the orders we computed with. Since $27 = 3^3$, P_{27} must be built over $\mathbb{F}_{27}$ and not over $\mathbb{Z}/27$. P_{27} is the unique most symmetric doubly regular tournament on 27 vertices, with $|\mathrm{Aut}(P_{27})| = 1053 = 3q(q-1)/2$, because the field automorphisms contribute (in general $|\mathrm{Aut}(P_q)| = q(q-1)e/2$ when $q = p^e$). It is still transitive on arcs and on non-arcs, which is what Corollary 5.2 needs. We ran $q = 27$ to completion and it is not 5-inducible; since that does not bear on $N(5)$ we report it here only as a runtime datapoint.

On a fixed base decomposition the total runtime *decreases* as the tournament grows, and so does the node count: a larger Paley tournament is *more* constrained, so the search is refuted nearer the root and the tree above each base state is smaller rather than larger. The work per node moves the other way, rising monotonically from 10.8 to 57.4 microseconds across the four runs below, because a larger tournament carries larger domains and each refinement step costs more. The node count falls faster than the per-node cost rises, which is what makes the total fall. The full runs:

Table G.2: The four full Paley sweeps.

q	live base states	core-hours	nodes	seconds per live base state
23	2,591	34.03	1.14×10^{10}	47.3
27	2,537	31.04	6.22×10^{9}	44.0
31	4,007	26.89	3.43×10^{9}	24.2
43	2,537	22.87	1.43×10^{9}	32.5

P_{43} is decided in *less time than* P_{27}, and explores 4.3 times fewer nodes to do it. The last column is the third divided by the second, so only the core-hours and the live base states are measured. Just two of these four runs share a decomposition, $q = 27$ and $q = 43$, on the same five-vertex base class and the same 2,537 live base states; $q = 23$ and $q = 31$ each used their own best configuration, and $q = 31$'s is the cheaper class of Appendix G.4, which is why its per-state figure is the lowest of the four. The core-hours fall monotonically in q either way. By contrast, P_{43} minus a vertex takes 185.5 core-hours, far more than P_{43} itself, as vertex deletion shrinks the automorphism group.

The search also gets faster as the margin is tightened. On the identical tournament ($P_{43} - v$), engine and 5-vertex base class, there is a factor of about 90. Unit margin admits 2,200 base states where majority admits 8,031; moreover, under exact margin the last voter is *forced*, since the support accumulated over the first $k - 1$ voters determines what the last must contribute, so a whole stage of the search collapses.

Two machines, and why their runtimes are not comparable. Some computations ran on a cluster rather than the laptop. Measured across six independent experiments, the cluster ran at 2 to 5 times the laptop's runtime per unit of work.

G.6 Runtimes for every computation

All runs in this paper ran on a single laptop (Apple M-series, 14 cores, 24 GB RAM), using at most 11 cores concurrently, with the following exceptions, which used a compute cluster: the census of regular tournaments on 15 vertices, the self-converse census of order 13, the order-12 census of Appendix A, six of the 22-vertex sweeps of Section 6.2, part of the certificate for $\Delta_1 - v_{10}$, 16 of the 19 unrestricted sweeps of doubly regular tournaments on 23 vertices in Section 6.7, and most of the search of Appendix C.

Runtimes are reported in *core-hours*, the workers' running times summed; cluster core-hours are not comparable with laptop ones (Appendix G.5), so we convert every cluster runtime to *laptop-equivalent* core-hours, as follows. Every core-hour figure in this paper is on the laptop's scale, except the two cluster figures that Section 8 marks as such.

The certificate for $\Delta_1 - v_{10}$ ran partly on each machine. Both machines solved 133 of its groups of five cubes, and on those the cluster took on average about 2.1 times as long as the laptop, between 1.6 and 3.1 times for individual groups; we divide its cluster time by that average. The factor is an average slowdown rather than an exact exchange rate, so the converted total is approximate. The searches of $\Delta_1 - v_1$, $\Delta_1 - v_2$, $\Delta_{11} - v_5$ and $\Delta_{11} - v_{20}$ also ran on the cluster, wholly or in part, and we convert them with a factor measured on these searches: 355 base states of $\Delta_{11} - v_5$ were searched on both machines, and the cluster took 3.55 times as long in total. The factor differs from the certificate's because the two computations run different programs. We apply the same factor to the other searches of this kind that ran on the cluster, the sweeps of $\Delta_6 - v_5$ and $\Delta_{28} - v_5$ and of 16 of the doubly regular tournaments on 23 vertices, for which no paired measurement exists. The self-converse census of order 13 ran partly on each machine, and there the cluster took 5.13 times as long per tournament, so we divide its cluster time by that factor. The censuses of orders 12 and 15 ran

on the cluster alone, so we measured their factors directly: we reran a random sample of their parts on the laptop, 8 of order 12 and 12 of order 15, with the same scripts, and the cluster took 3.85 and 4.28 times as long respectively. Table G.3 lists the result.

Table G.3: The runtime of every computation in this paper.

computation	margin	verdict	core-hours
P_{19}, certified refutation	unit	not inducible	25.1
P_{23}, certified refutation	unrestricted	not inducible	236.0
$\Delta_1 - v_{10}$, sweep (Corollary 5.5)	unrestricted	not inducible	96.9
$\Delta_1 - v_{10}$, certified refutation	unrestricted	not inducible	732
$\Delta_6 - v_5$, certified refutation	unrestricted	not inducible	1,292.7
$\Delta_6 - v_5$, sweep	unrestricted	not inducible	437.0
$\Delta_{28} - v_5$, sweep	unrestricted	not inducible	433.2
16 doubly regular on 23 vertices	unrestricted	not inducible	7,458.2
$\Delta_1 - v_1$, sweep (Corollary 5.5)	unrestricted	not inducible	136.8
$\Delta_1 - v_2$, sweep (Corollary 5.5)	unrestricted	not inducible	122.9
$\Delta_{11} - v_2$, sweep (Corollary 5.5)	unrestricted	not inducible	108.5
$\Delta_{11} - v_5$, sweep (Corollary 5.5)	unrestricted	not inducible	124.2
$\Delta_{11} - v_{20}$, sweep (Corollary 5.5)	unrestricted	not inducible	93.6
$P_{31} - v$, sweep	unrestricted	not inducible	239.5
P_{23}, one arc reversed	unrestricted	inducible	210.4
second doubly regular 19, all 57 arc reversals	unit	all inducible	3.59
regular on 15 vertices	unit	all inducible	726.1
self-converse on 13 vertices	unit	all inducible	294.2
all on 12 vertices	unit	all inducible	2,615.6

The four full Paley sweeps, $q = 23$, 27, 31 and 43, are tabulated with their node counts in Appendix G.5.

Appendix H. The obstructions on 22 vertices

This appendix gives the obstructions of Sections 6.2 and 6.3 so that each can be checked without the catalogue of McKay (2026). Each is given in the catalogue's own format, a single string of 231 bits giving the upper triangle of its adjacency matrix row by row: for every $u < w$, in the labelling of the parent with the deleted vertex removed, 1 if the u-th vertex beats the w-th and 0 otherwise. Each string is printed in three lines of 77, below the order of the automorphism group and, when it is non-trivial, a minimal set of generators in cycle notation, in the same labels.

$\Delta_1 - v_1$ $\quad$ |Aut| = 5, generated by (2 9 11 7 5)(3 8 4 12 6)(13 22 15 16 20)(14 17 19 23 18)

```
11111000001111110000001100011110001111100100111000010110011111100110110010101
11011010101100110101011100101100110100110000101110010101101101010101100011100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_1 - v_2$ $\quad$ |Aut| = 5, generated by (1 19 12 23 8)(3 6 4 18 14)(5 16 15 13 7)(9 22 11 20 21)

```
11111111110000000000001100011110001111100100111000010110011111100110110010101
11011010101100110101011100101100110100110000101110010101101101010101100011100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_1 - v_{10}$ $\quad$ |Aut| = 55, generated by (2 9 11 7 5)(3 8 4 12 6)(13 22 15 16 20)(14 17 19 23 18) and (1 18 4 14 23 19 12 6 8 17 3)(2 16 21 7 13 5 11 9 22 15 20)

```
11111111110000000000011111000011111100000011000111000111110010011000010110011
11110110110010101110101010110011010111100101100110001100001011101010110110100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_{11} - v_2$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
111111111100000000000011101100011000011110010101100111010011101011000101111 00
111010110010100111110110101110110001010111001110001010110010110101100011101 01
110010010111000111010100010111100100110010101010001010101100100010111011101 11
```

$\Delta_{11} - v_5$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
111111111100000000000111100000111111000000110110001100001111010101100111010 01
111010110010100111110110101110110001010111001110001010110010110101100011101 01
110010010111000111010100010111100100110010101010001010101100100010111011101 11
```

$\Delta_{11} - v_{20}$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
111111111110000000000111110000011111100000011101100011000011100101011001110101
110101100010111001110101100101011111011010111010001010111001100010101100111 01
011000110101110010001110001111010001011100100100101011000100101101001011011 11
```

$\Delta_6 - v_5$ $|\mathrm{Aut}| = 1$

```
111111111100000000000111100000111111000000110011010010011110011000100001111 01
110111000101101101111111101111000010101100100101110101010101110100110101011 01
101100110011010011010110001100110110101010001110010010101001100100111001011 10
```

$\Delta_{28} - v_5$ $|\mathrm{Aut}| = 1$

```
111111111100000000000111100000111111000000111010010001010111010101000010111 01
111001100101101110111111101111000001101101000111000111000101011101010101100 11
011010101100011011010110001101010110110100001110001001100000110100111001011 10
```

$\Delta_1 - v_{10}$ **with the arc between** v_2 **and** v_3 **reversed** $|\mathrm{Aut}| = 1$

```
111111111100000000000011110000111111000000110001110001111100100110000101101 11
111101101100101011101010101100110101111001011001100011000010111010101101101 00
101010110010111011010011000101101010110101011110000011010100010001010111100 11
```

Appendix I. Criticality at 21 vertices

This appendix carries the detail behind Section 6.6. Four of the 110 vertex-transitive tournaments on 21 vertices are unit-margin obstructions (Section 6.7). We name them by their automorphism groups, all of order 21: C_a and C_b have the cyclic group $\mathbb{Z}_{21}$, and F_a and F_b the nonabelian group $\mathbb{Z}_7 \rtimes \mathbb{Z}_3$.

Vertex-critical but maximally not arc-critical. Arc-criticality implies vertex-criticality, and the converse fails here as widely as it can. Each of the four is *vertex-critical* at unit margin, and for C_a every single-arc reversal remains a unit-margin obstruction.

Arc-criticality is a property of the arc, not just the tournament. All four have ten arc orbits of size 21; we sweep one representative per orbit at unit margin and get results that differ by tournament *and* by arc, `U` marking a reversal that is still not unit-margin inducible and `S` one that is. C_b splits five orbits to five, so exactly half its arcs are critical and we call it *arc-semi-critical*; with P_{23} arc-critical unrestricted and C_a critical nowhere, all three possibilities occur among our tournaments. Every one of the 15 reversals that remains an obstruction belongs to a tournament with *cyclic automorphism group*, while all 20 orbits of the two with the nonabelian group of order 21 are inducible.

Table I.1: Single-arc reversals of the four vertex-transitive unit-margin obstructions on 21 vertices, one per arc orbit.

tournament	Aut	the ten orbits	obstructions
C_a	$\mathbb{Z}_{21}$	U U U U U U U U U U	10
C_b	$\mathbb{Z}_{21}$	U S U S S U U S S U	5
F_a, F_b	$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	S S S S S S S S S S	0

Appendix J. Locally transitive tournaments, at unit margin, for every odd $k \geq 3$

A tournament is *locally transitive* if for every vertex v both $N^+(v)$ and $N^-(v)$ induce transitive subtournaments. Equivalently it is *round*: the vertices admit a cyclic order $v_0, \ldots, v_{n-1}$ in which every $N^-(v) \cup \{v\}$ and $\{v\} \cup N^+(v)$ is an interval, so that, orienting the cyclic order along the arcs, each vertex beats exactly the $d_i = |N^+(v_i)|$ vertices that follow it: $N^+(v_i) = \{v_{i+1}, \ldots, v_{i+d_i}\}$, indices modulo n. The equivalence is due to Brouwer (1980), who derived the cyclic order on the way to enumerating the class, and to Alspach and Tabib (1982), who state it for tournaments under the name *domination orientable*.

Theorem J.1. *Every locally transitive tournament is* 3*-inducible with unit margin, and hence* k*-inducible with unit margin for every odd* $k \geq 3$*.*

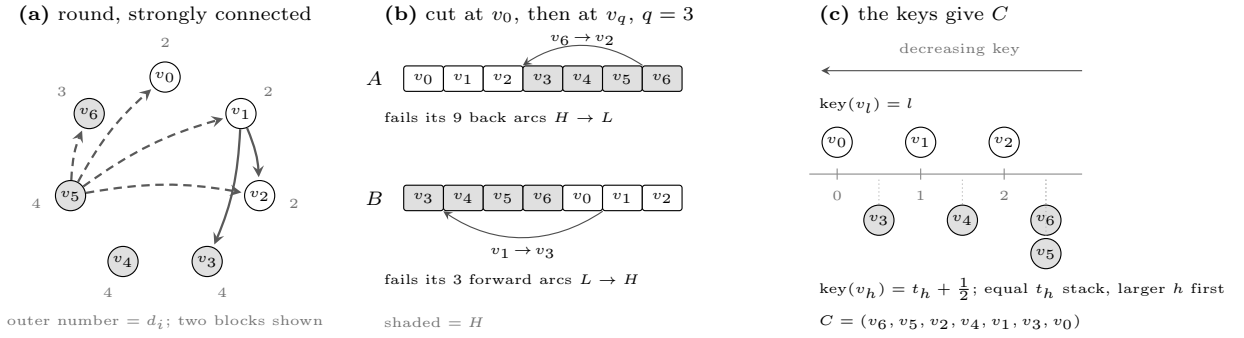


Figure J.1: The construction of Appendix J on a round tournament with out-degree sequence $(2, 2, 2, 4, 4, 4, 3)$, for which $q = 3$. **(a)** The cyclic order, each v_i labelled with its out-degree d_i, so that its out-neighbours are $v_{i+1}, \ldots, v_{i+d_i}$; the two blocks drawn are those of $v_1 \in L$ and $v_5 \in H$, and that of v_5 wraps, with back-targets v_0, v_1, v_2. **(b)** A and B are the same cyclic order cut at v_0 and at v_q, each failing exactly the arcs that cross its own cut, 9 back arcs and 3 forward arcs; one of each is drawn. **(c)** The keys, C being the vertices in decreasing order of key. A vertex of L sits at its index and one of H half a unit past its ceiling, so reading leftwards puts v_h before v_l exactly when $v_h \to v_l$; v_5 and v_6 share a ceiling and the tie-break separates them, giving $C = (v_6, v_5, v_2, v_4, v_1, v_3, v_0)$.

A locally transitive tournament that is not strongly connected is transitive: a strongly connected component C with $|C| \geq 3$ contains a directed triangle, and any vertex outside C either beats all of C or loses to all of it, contradicting local transitivity. A transitive tournament is 1-inducible at margin 1, its single voter supporting every arc. Adding an order together with its reverse adds one supporter and one opponent to each arc, raising every support by 1 and k by 2 and so leaving every margin $2c(e) - k$ unchanged. Padding with the identity order $(1, 2, \ldots, n)$ and its reverse therefore gives unit margin at $k = 3$, and repeating gives every odd k.

Let T be strongly connected and round, and let d_i be the out-degree of vertex i; Figure J.1 illustrates what follows, on seven vertices. Cutting the cyclic order at v_0 gives the linear order $A = (v_0, \ldots, v_{n-1})$. We write $r_i = i + d_i$, an integer deliberately *not* reduced modulo n. For each v_i, the out-neighbours are then $v_{i+1}, \ldots, v_{r_i}$ with indices reduced only when they are read as vertices. The sequence r is non-decreasing, since if $r_{i+1} < r_i$ then $v_{r_{i+1}+1}$ is an out-neighbour of v_i, while its own interval includes v_i, a contradiction. Let $H = \{i : r_i \geq n\}$ be the vertices with an out-neighbour that the cut defined by A placed *before* them. As

r is non-decreasing, $H = \{q, \dots, n-1\}$ for some q; write $L = \{0, \dots, q-1\}$. Neither set is empty: $H = \emptyset$ would make v_{n-1} a sink and $L = \emptyset$ would force $d_0 \geq n$, so neither occurs. For $h \in H$ put $t_h = r_h - n$, so that the out-neighbours of v_h placed before it by the cut are exactly $v_0, \dots, v_{t_h}$: we call them the *back-targets* of v_h, and t_h its *ceiling*. By monotonicity of r, $t_q \leq \cdots \leq t_{n-1}$. Every back-target lies in L, that is $t_{n-1} < q$: otherwise $v_{n-1} \to v_q$, while $r_q \geq n$ implies $v_q \to v_{n-1}$. The arcs between L and H therefore divide into the *back arcs* going from H to L, and the *forward arcs* going from L to H; and for $h \in H$ and $l \in L$ we have $v_h \to v_l$ exactly when $l \leq t_h$.

We give each vertex a *key*: $\mathrm{key}(v_l) = l$ for $l \in L$ and $\mathrm{key}(v_h) = t_h + \frac{1}{2}$ for $h \in H$, ties among equal ceilings broken in favour of the larger index, and let C list the vertices in decreasing order of key. Integer and half-integer keys never collide, so that tie-break is the only one needed, the keys are pairwise distinct and C is a linear order; and for $h \in H$ and $l \in L$, $\mathrm{key}(v_h) > \mathrm{key}(v_l) \iff t_h + \frac{1}{2} > l \iff l \leq t_h \iff v_h \to v_l$, so C places v_h before v_l exactly when $v_h \to v_l$. We now take the three orders $A = (v_0, \dots, v_{n-1})$, $B = (v_q, \dots, v_{n-1}, v_0, \dots, v_{q-1})$, which is the same cyclic order cut at v_q instead of at v_0, and C. Every arc now has support exactly two. *Arcs with both endpoints inside L or inside H* are only supported by A and B; *forward arcs* $v_l \to v_h$ are only supported by A and C; and *back arcs* $v_h \to v_l$ are only supported by B and C, so the profile realizes T at unit margin. The padding argument above extends this margin to any odd $k \geq 3$. ■